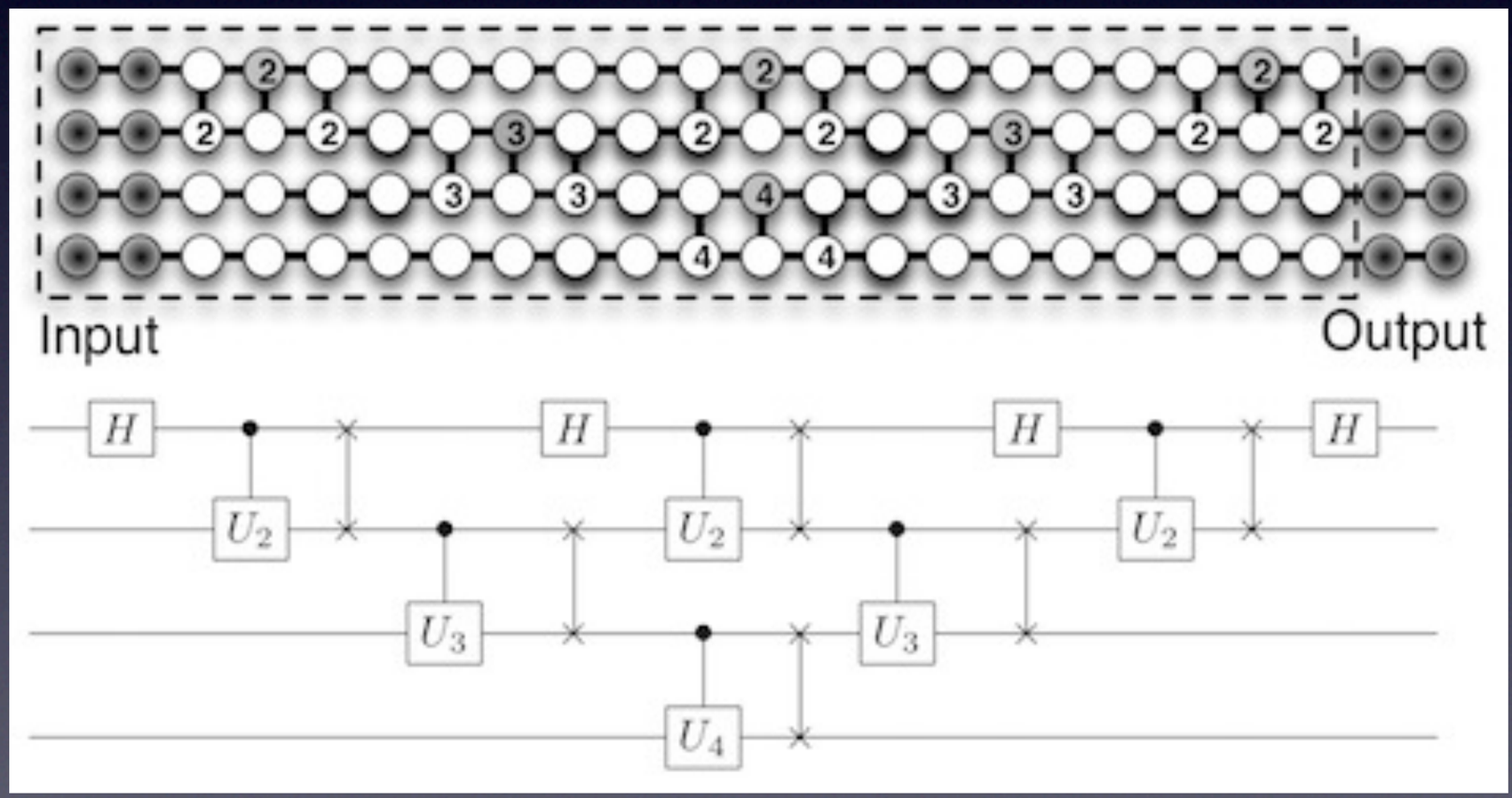
Adiabatic Quantum Transistors

Dave Bacon
University of Washington



DARPA QuEST Grant: "Robust Quantum Gadgets"

Experimentalists...

Do you have qubits with...

- exchange interactions?

$$X \otimes X + Y \otimes Y + Z \otimes Z$$

- anisotropic exchange interactions?

$$X \otimes X + Y \otimes Y$$

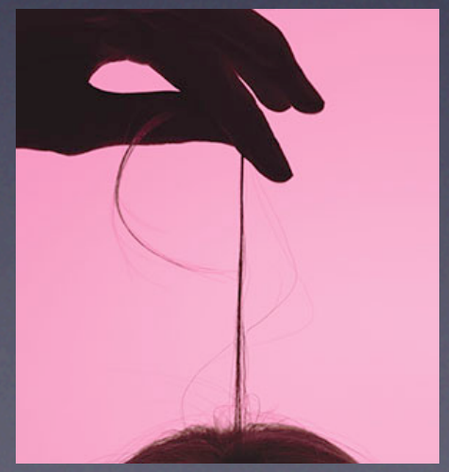
- twisted exchange interactions?

$$X \otimes Y + Y \otimes X$$

Do you tear your hair out dealing with timing errors?

THIS TALK IS FOR YOU

(but first a word from our sponsors)



Robust Quantum Gadgets

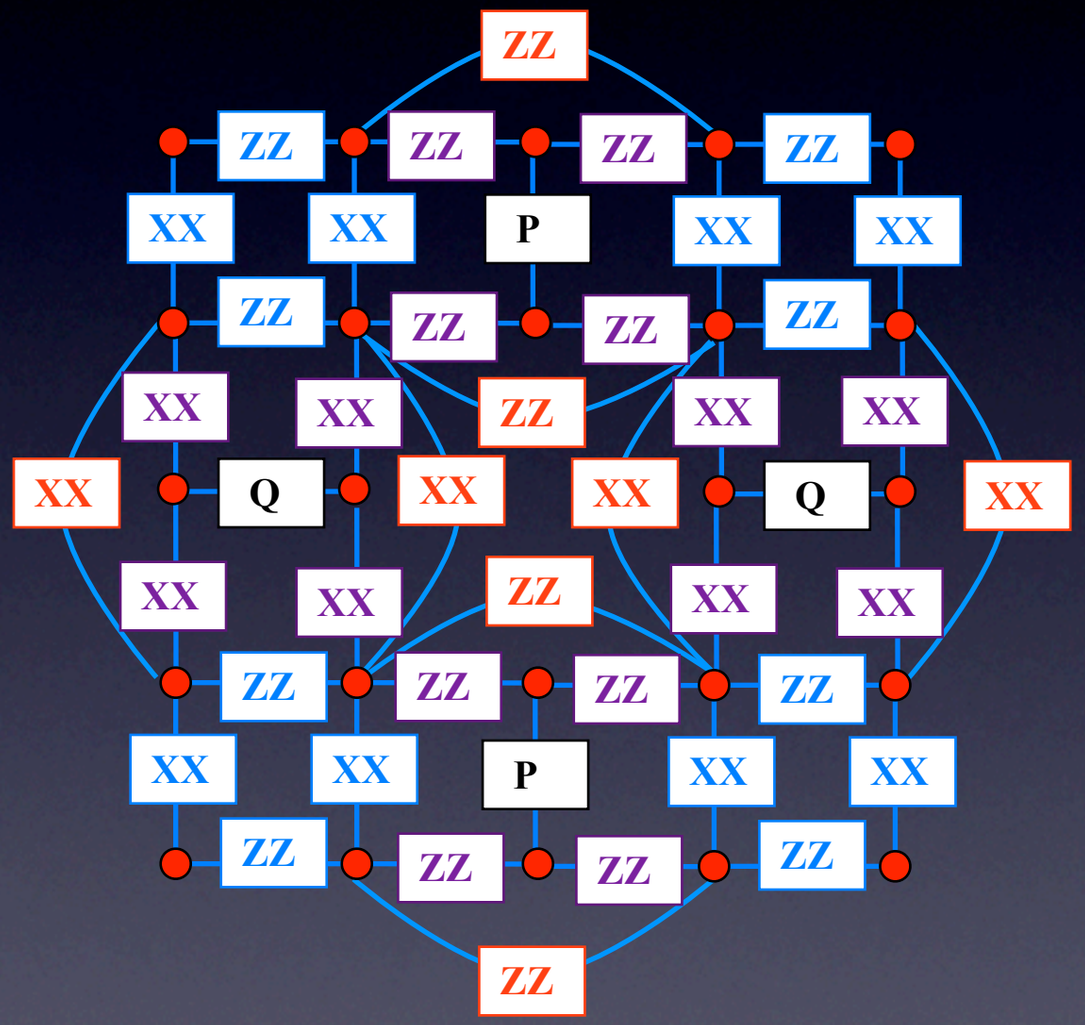
Quantum error correction via designer Hamiltonians

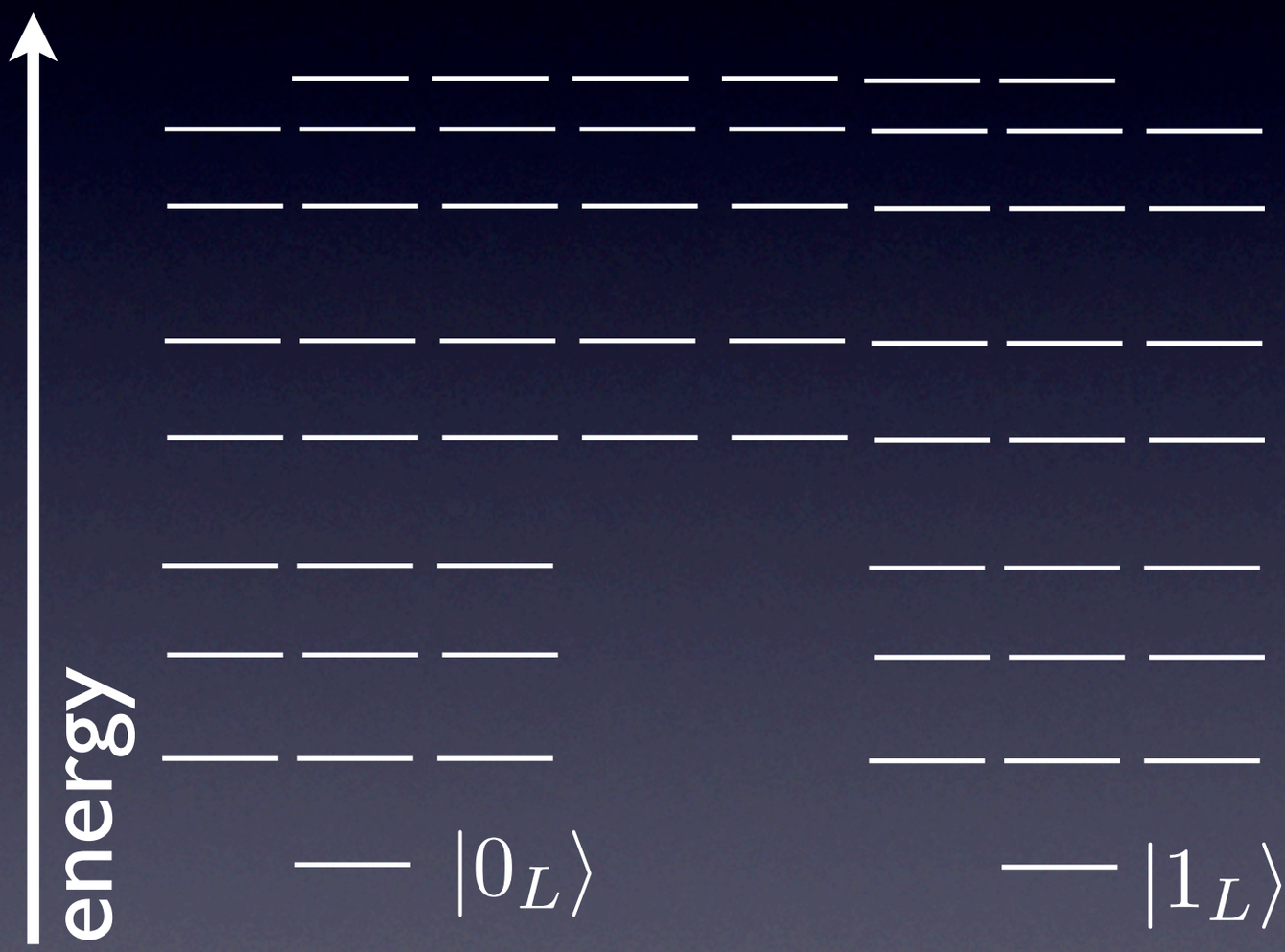
- Amplifying gadgets (see Poster)
- Fault-tolerant bang-bang stabilizer Hamiltonians (see Poster)
- CodeQuest (see Poster or Gregory Crosswhite)

Fault-tolerance of universal adiabatic quantum computing

- Successfully achieved FT for classical analogue (See Poster)

Protecting Quantum Information





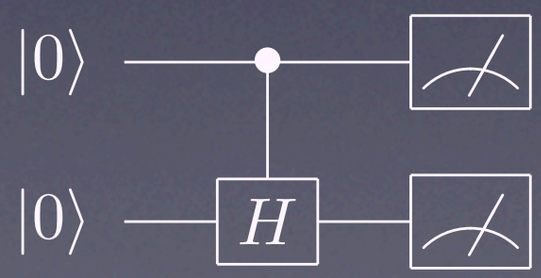
But...how to compute on encoded quantum information?

Topological QC

[Kitaev, Annals Phys. 303, 2 (2003), preprint (1997)]

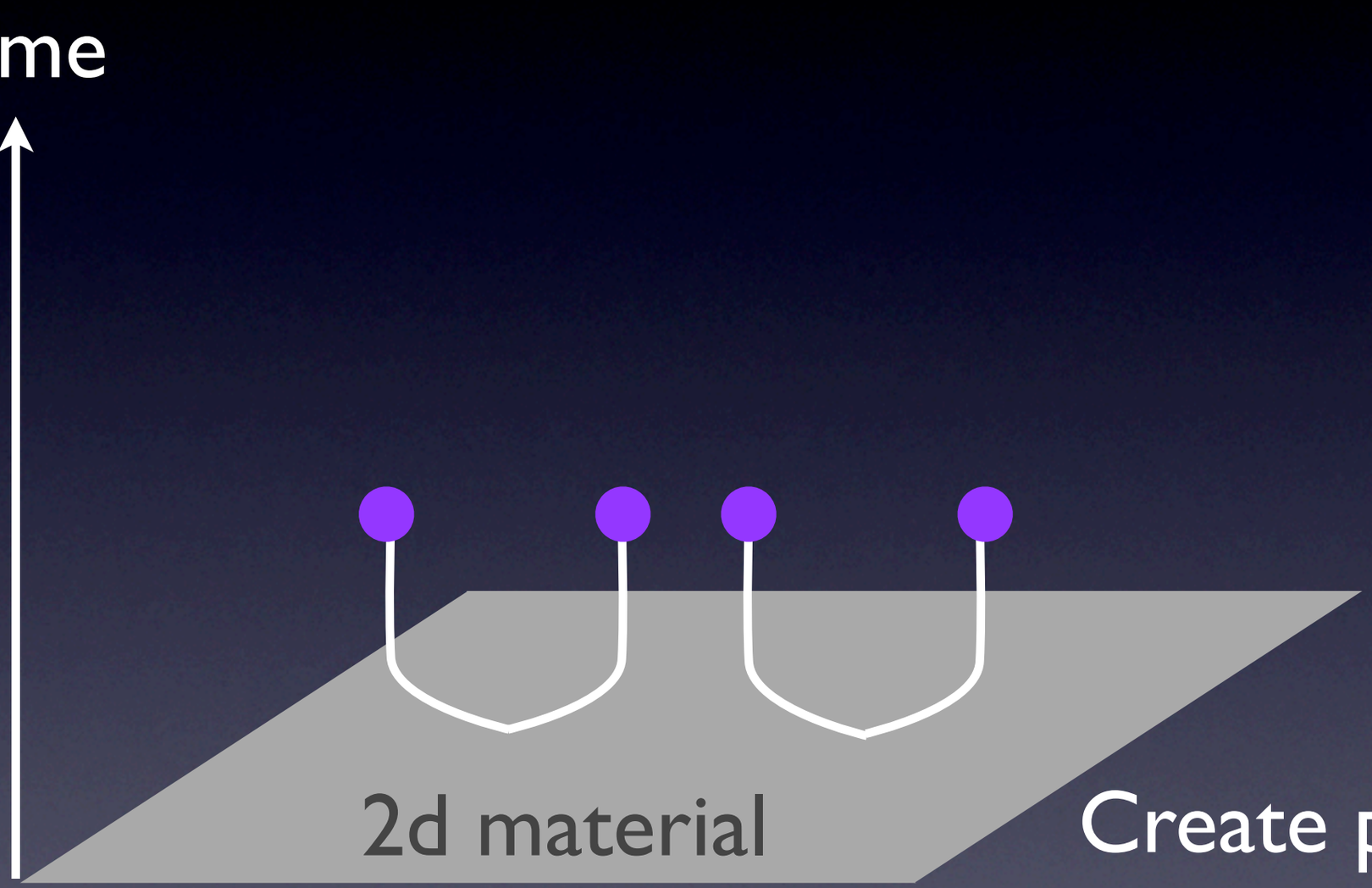
Time

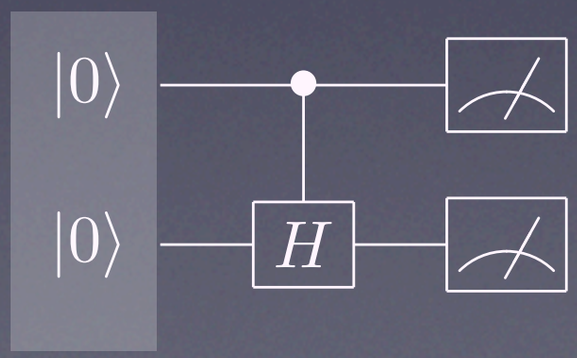
2d material



Topological QC

[Kitaev, Annals Phys. 303, 2 (2003), preprint (1997)]

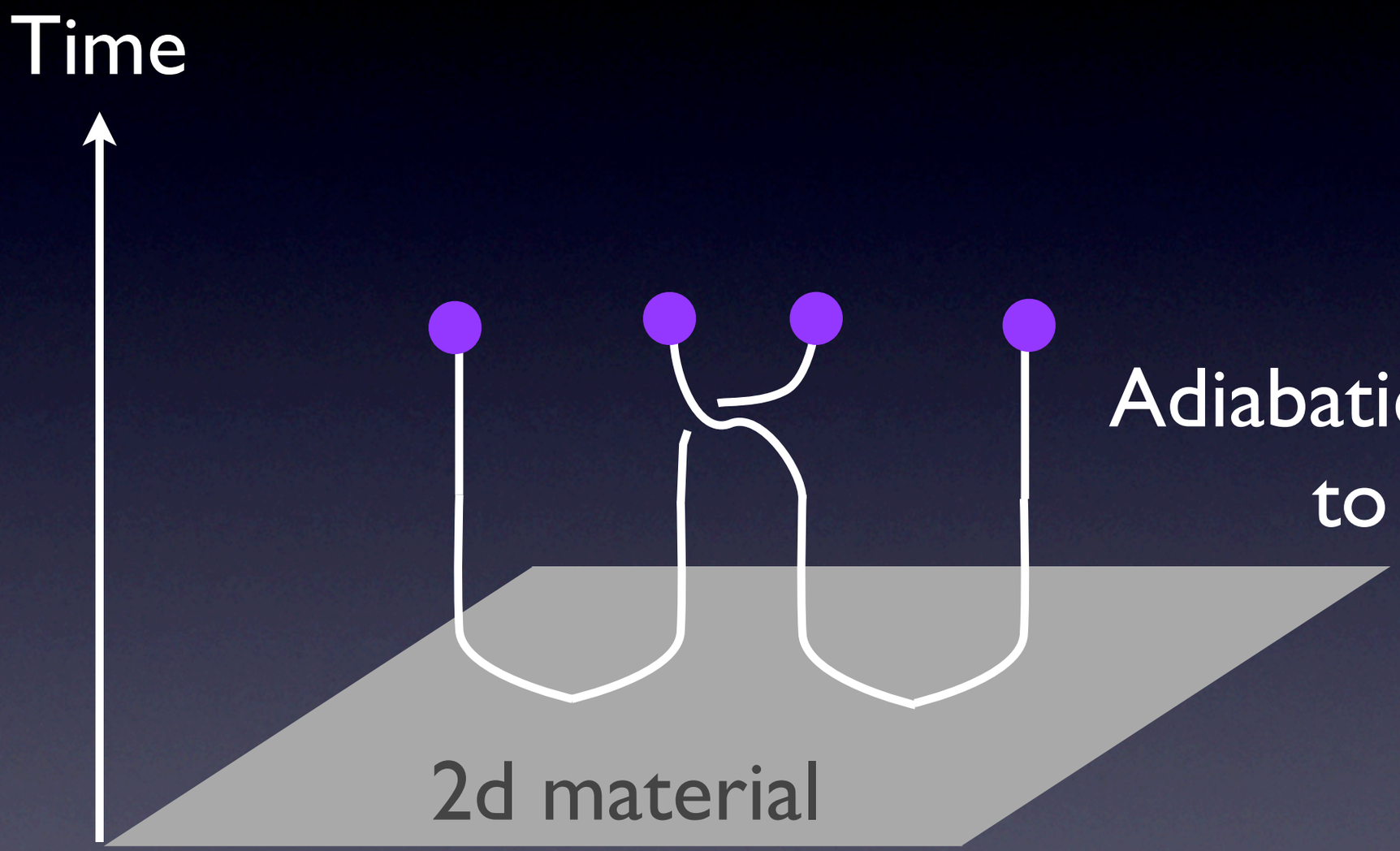




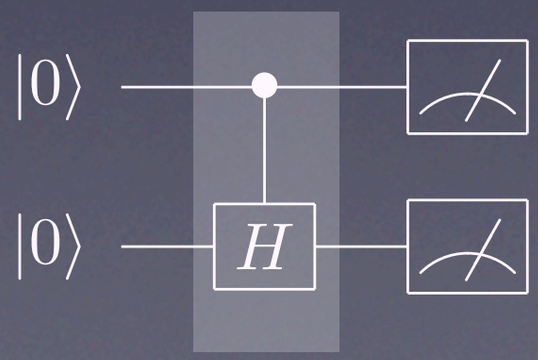
Create particles (anyons) out of ground state

Topological QC

[Kitaev, Annals Phys. 303, 2 (2003), preprint (1997)]

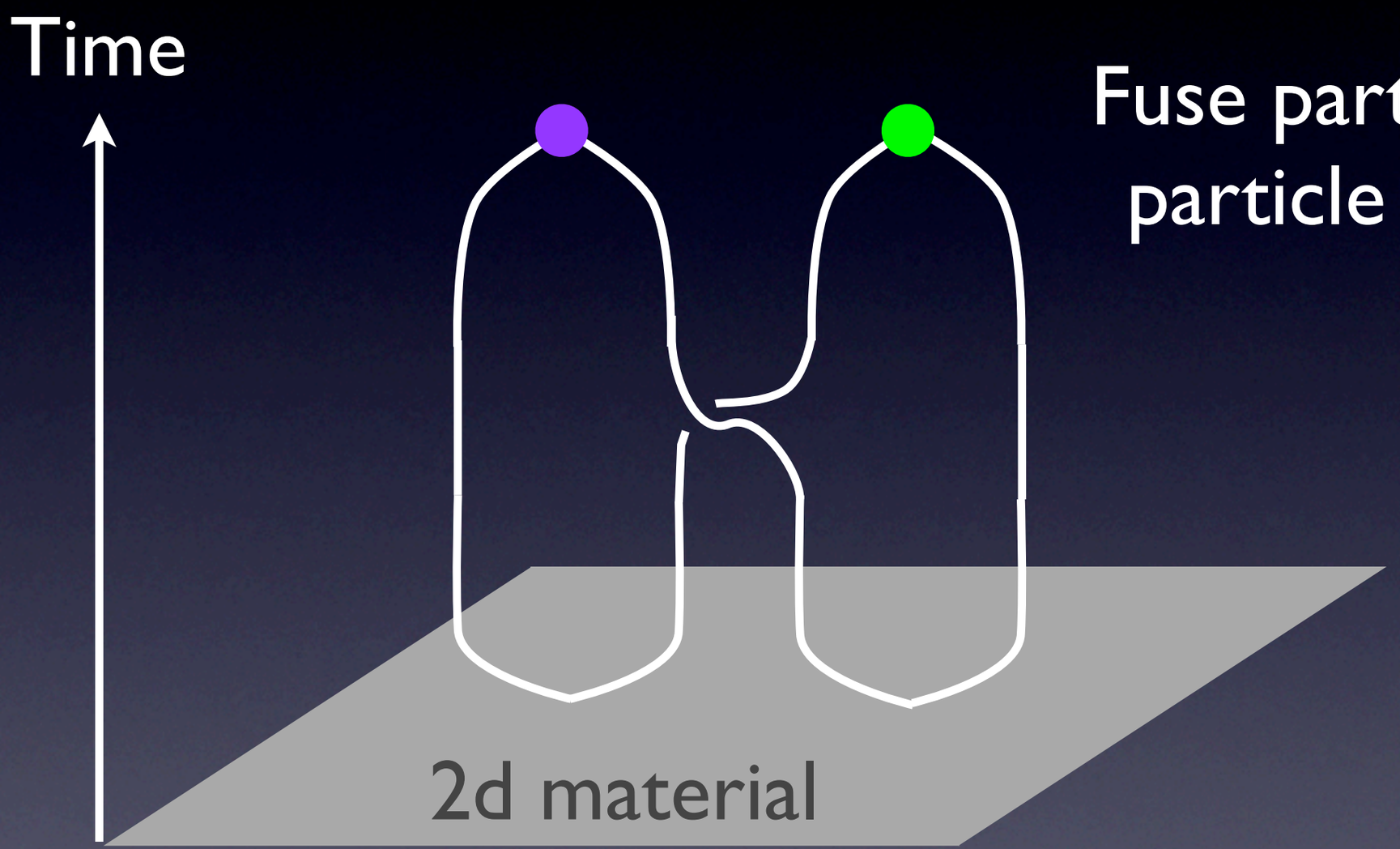


Adiabatically braid anyons to enact gates

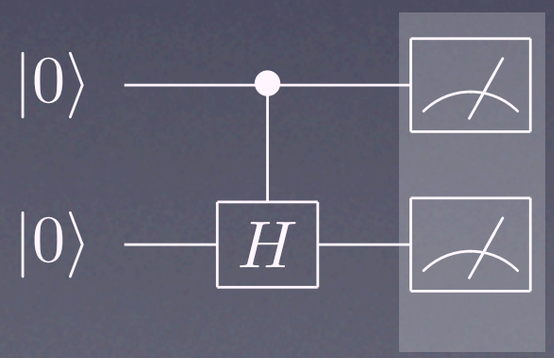


Topological QC

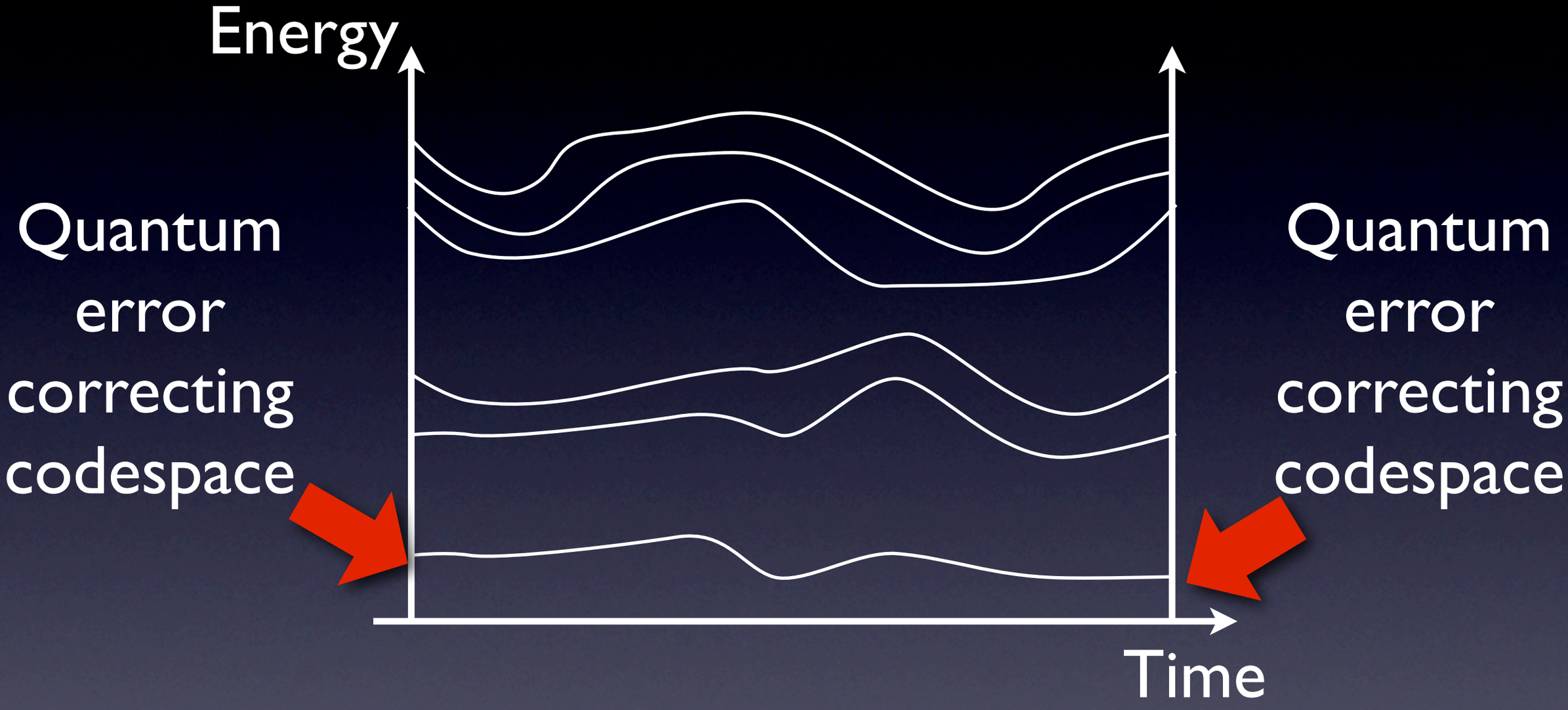
[Kitaev, Annals Phys. 303, 2 (2003), preprint (1997)]



Fuse particles and observe particle type to measure



Adiabatic Dragging



Require: degenerate through entire evolution

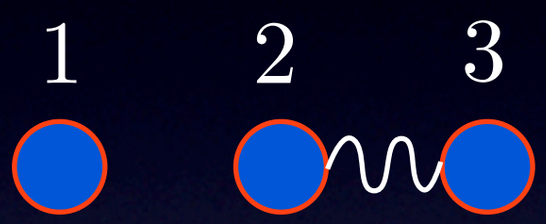
"Open loop" holonomic evolution

[D. Kult, J. Aberg, E. Sjoqvist, Phys. Rev. A 74, 022106 (2006)]

Two Simple "Codes"

both on three qubits

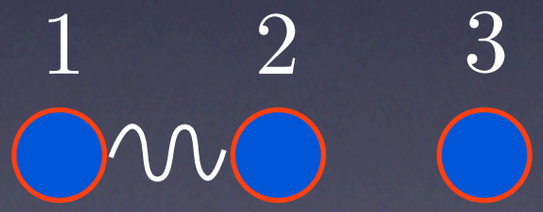
Code 1



Common +1 eigenspace of $\begin{cases} X_2X_3 \\ Z_2Z_3 \end{cases}$

$$(\alpha|0\rangle + \beta|1\rangle) \otimes |\Phi_+\rangle \qquad |\Phi_+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Code 2



Common +1 eigenspace of $\begin{cases} X_1X_2 \\ Z_1Z_2 \end{cases}$

$$|\Phi_+\rangle \otimes (\alpha'|0\rangle + \beta'|1\rangle)$$

Code ⇒ Hamiltonian



$$H_1 = -\Delta(X_2X_3 + Z_2Z_3) \qquad \Delta > 0$$

ground states: $|0\rangle \otimes |\Phi_+\rangle$, $|1\rangle \otimes |\Phi_+\rangle$



$$H_2 = -\Delta(X_1X_2 + Z_1Z_2) \qquad \Delta > 0$$

ground states: $|\Phi_+\rangle \otimes |0\rangle$, $|\Phi_+\rangle \otimes |1\rangle$

Code ⇒ Hamiltonian



$$H_1 = -\Delta(X_2X_3 + Z_2Z_3) \qquad \Delta > 0$$

ground states: $|0\rangle \otimes |\Phi_+\rangle$, $|1\rangle \otimes |\Phi_+\rangle$



$$H_2 = -\Delta(X_1X_2 + Z_1Z_2) \qquad \Delta > 0$$

ground states: $|\Phi_+\rangle \otimes |0\rangle$, $|\Phi_+\rangle \otimes |1\rangle$

What happens if we (adiabatically) turn off H_1 while turning on H_2?

$$H(t) = \left(1 - \frac{t}{T}\right) H_i + \frac{t}{T} H_f$$



What Happens?

$$\bar{X}_1 = X_1X_2 \qquad \bar{Z}_1 = Z_2Z_3$$
$$\bar{X}_2 = X_2X_3 \qquad \bar{Z}_2 = Z_1Z_2$$
$$\bar{X}_3 = X_1X_2X_3 \qquad \bar{Z}_3 = Z_1Z_2Z_3$$

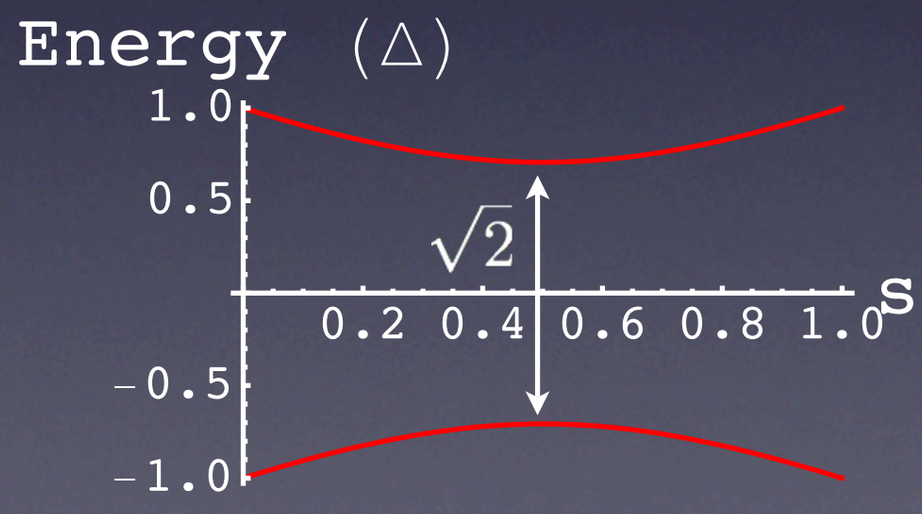
Initial Hamiltonian

Final Hamiltonian

$$H_i = -\Delta(\bar{Z}_1 + \bar{X}_2)$$

$$H_f = -\Delta(\bar{X}_1 + \bar{Z}_2)$$

$$H(s) = (1-s)H_i + sH_f$$



Adiabatic Teleportation

$$\begin{aligned} \bar{X}_1 &= X_1 X_2 \\ \bar{X}_2 &= X_2 X_3 \\ \bar{X}_3 &= X_1 X_2 X_3 \end{aligned} \qquad \begin{aligned} \bar{Z}_1 &= Z_2 Z_3 \\ \bar{Z}_2 &= Z_1 Z_2 \\ \bar{Z}_3 &= Z_1 Z_2 Z_3 \end{aligned}$$

Initial Hamiltonian

$$H_i = -\Delta(\bar{Z}_1 + \bar{X}_2)$$

Info in first qubit:

$$X_1 = \bar{X}_2 \bar{X}_3$$
$$Z_1 = \bar{Z}_1 \bar{Z}_3$$

Final Hamiltonian

$$H_f = -\Delta(\bar{X}_1 + \bar{Z}_2)$$

Info in second qubit:

$$X_3 = \bar{X}_1 \bar{X}_3$$
$$Z_3 = \bar{Z}_2 \bar{Z}_3$$

$$|\psi\rangle \otimes |\Phi_+\rangle \quad \rightarrow \quad |\Phi_+\rangle \otimes |\psi\rangle$$

Adiabatic Teleportation

[Bacon and Flammia, Phys. Rev. Lett. 103, 120504 (2009)]



Adiabatic (time scale set by strength of interaction)

Works for multitude of other interactions, for example:

$$H_{ex} = \Delta(X_iX_j + Y_iY_j + Z_iZ_j)$$

$$H_{ani} = -\Delta(X_iX_j \pm Y_iY_j)$$

$$\Delta > 0$$

Errors arise from uncontrolled interactions which break the symmetry of the logical qubits

Adiabatic Gate Teleportation

[Bacon and Flammia, Phys. Rev. Lett. 103, 120504 (2009)]

$$H_i = -U_3\Delta(X_2X_3 + Z_2Z_3)U_3^\dagger \qquad H_f = -\Delta(X_1X_2 + Z_1Z_2)$$

Note: spectrum unchanged by unitary

$$|\psi\rangle \otimes |\Phi_+\rangle \longrightarrow |\Phi_+\rangle \otimes U_3|\psi\rangle$$

Example (Hadamard gate): $H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$$H_i = -\Delta(X_2Z_3 + Z_2X_3) \qquad H_f = -\Delta(X_1X_2 + Z_1Z_2)$$

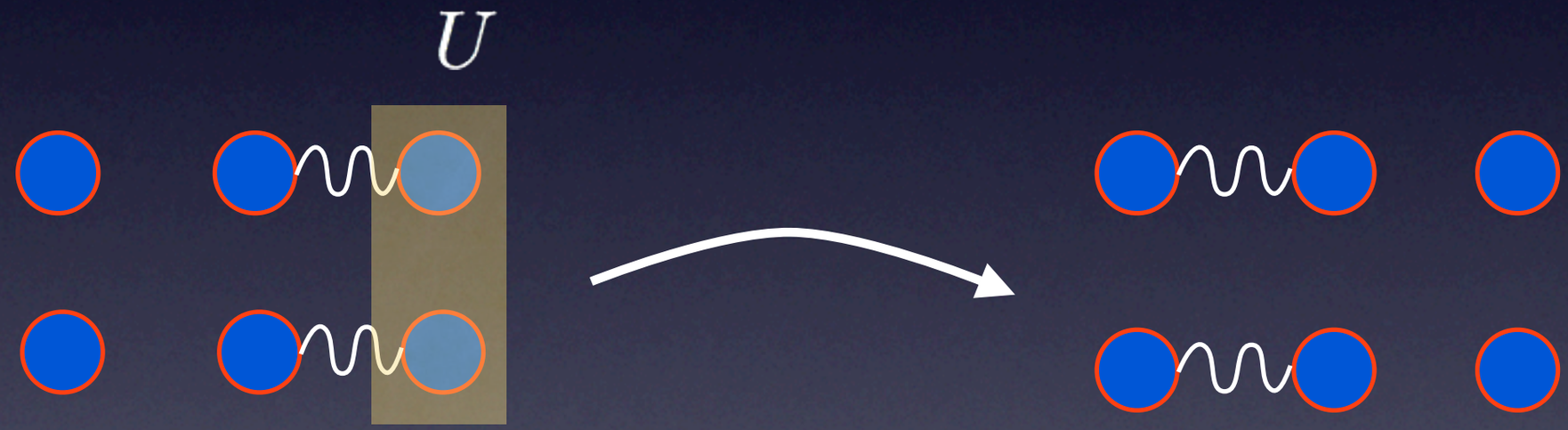
$$|\psi\rangle \otimes |\Phi_+\rangle \longrightarrow |\Phi_+\rangle \otimes H|\psi\rangle$$

Let's Build a Computer

One qubit gates



Two qubit gates



Problem: initial Hamiltonian requires 3 qubit interactions

Example



$$H_i = -\Delta(X_2X_3Z_6 + Z_2Z_3 + Z_5Z_6 + Z_3X_5X_6)$$

To get 2 qubit interactions: perturbation theory gadgets

Gadgets





Replace 3,6 by encoded qubits: $|0_L\rangle = |00\rangle, |1_L\rangle = |11\rangle$

$$
\begin{aligned}
H_i &= -\lambda(X_2X_{3a} + Z_2Z_{3a}) - \lambda(X_5X_{6a} + Z_5Z_{6a}) \\
&\quad -\omega Z_{3a}Z_{3b} - \lambda(X_{3b}Z_{6b} + Z_{3b}X_{6b}) \qquad \lambda \ll \omega \\
H_f &= -\lambda(X_1X_2 + Z_1Z_2) - \lambda(X_4X_5 + Z_4Z_5) \\
&\quad -\omega Z_{3a}Z_{3b}
\end{aligned}
$$

[S. D. Bartlett and T. Rudolph, Phys. Rev. A 74, 040302 (2006)]

Gadgets



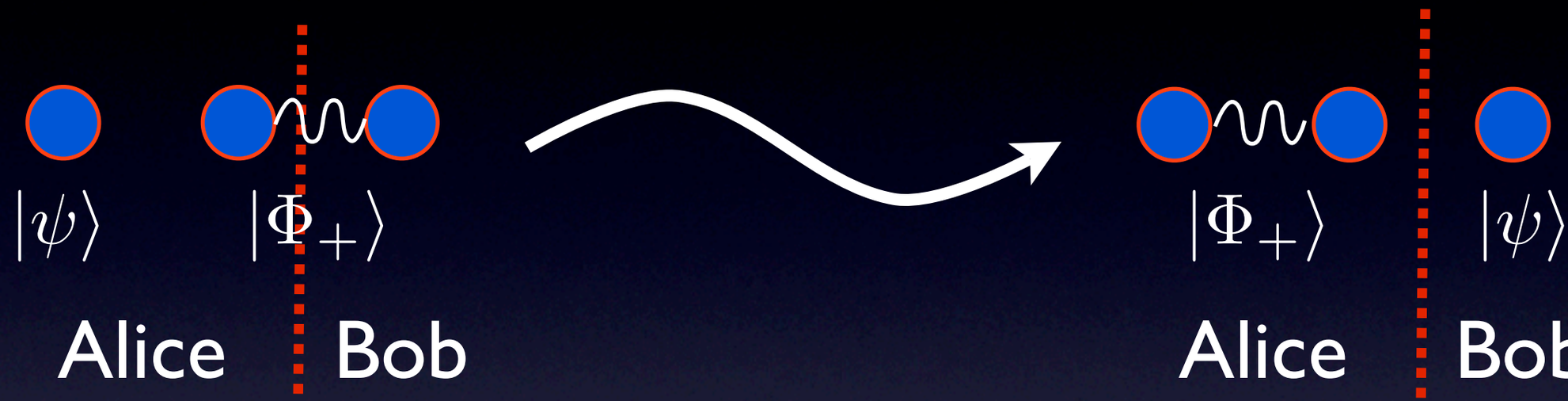


$$H_i = -\lambda(X_2X_{3a} + Z_2Z_{3a}) - \lambda(X_5X_{6a} + Z_5Z_{6a})$$
$$-\omega Z_{3a}Z_{3b} - \lambda(X_{3b}Z_{6b} + Z_{3b}X_{6b})$$
$$H_f = -\lambda(X_1X_2 + Z_1Z_2) - \lambda(X_4X_5 + Z_4Z_5)$$
$$-\omega Z_{3a}Z_{3b}$$

Minimum gap: $O\left(\frac{\lambda^2}{\omega}\right)$

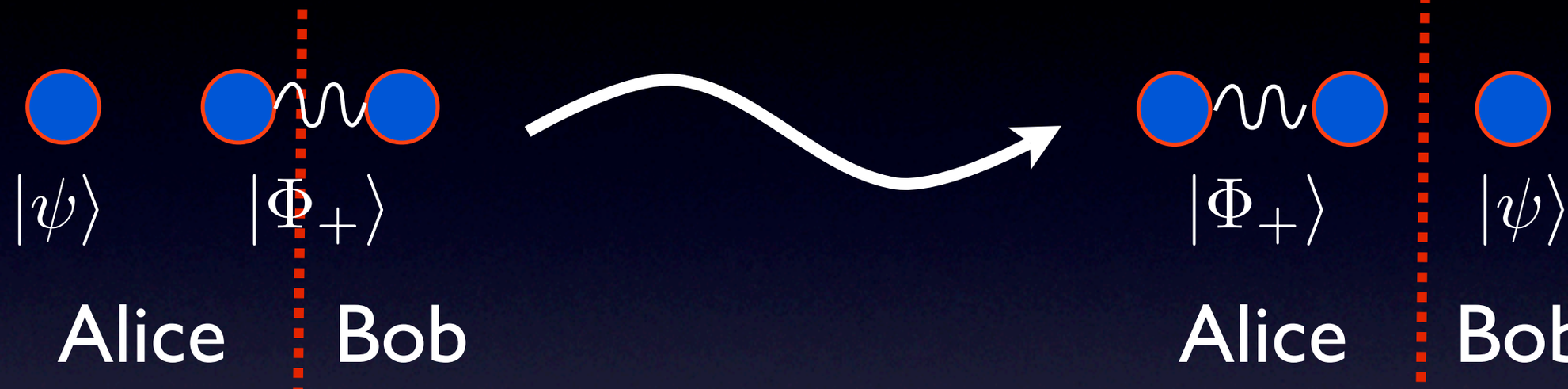
Initial ground state fidelity: $1 - O\left(\frac{\lambda^2}{\omega^2}\right)$

Anything You Can Do...



Like teleportation, but we have *post-selected* Alice's Bell measurement result

Anything You Can Do...





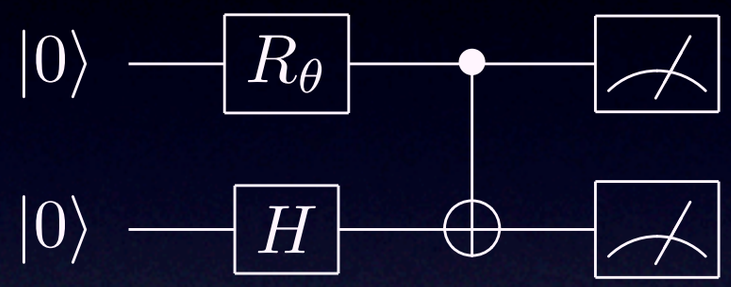
Like teleportation, but we have *post-selected* Alice's Bell measurement result

"Anything you can do, I can do adiabatically, I can do anything slower than you..."

One Way QC



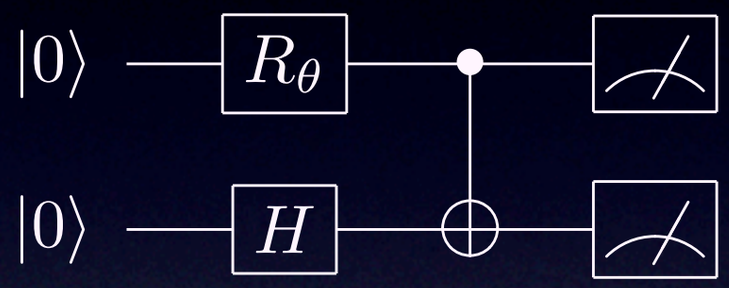
[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]

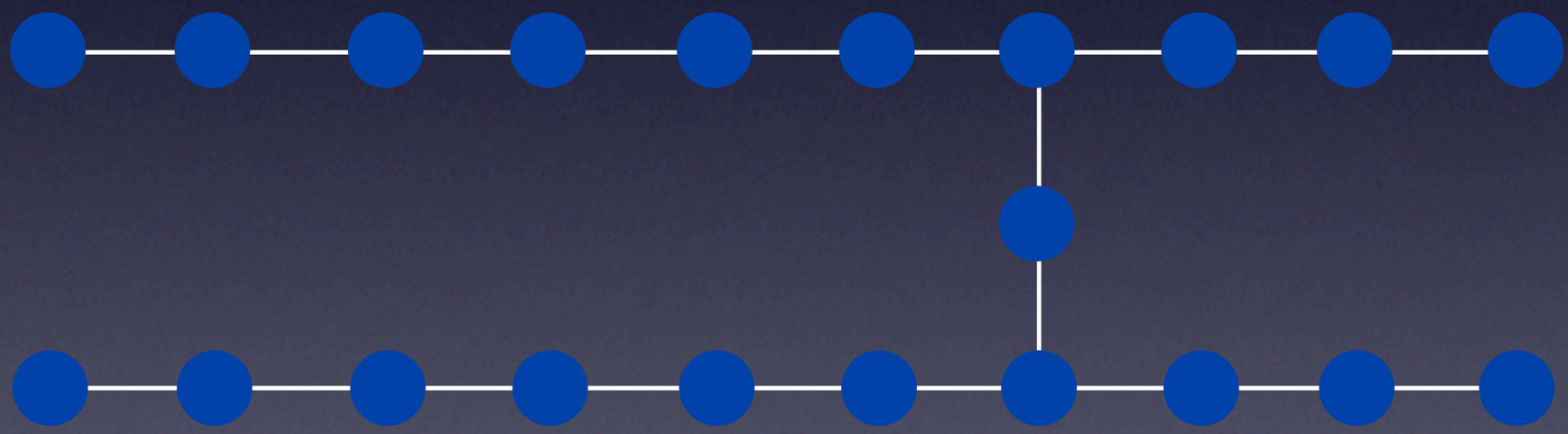


One Way QC



[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]



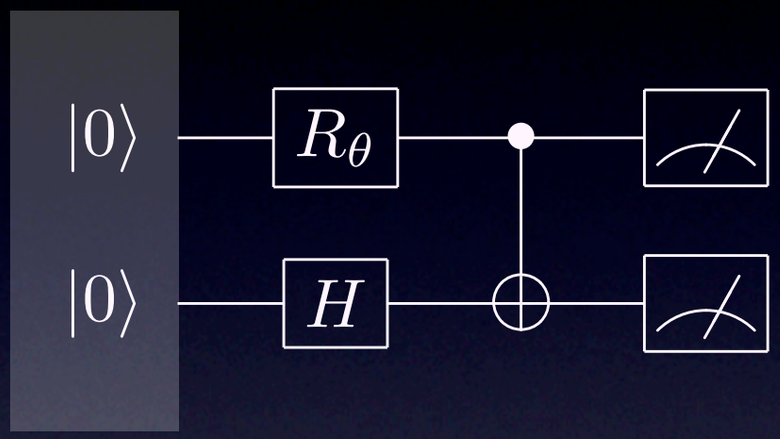


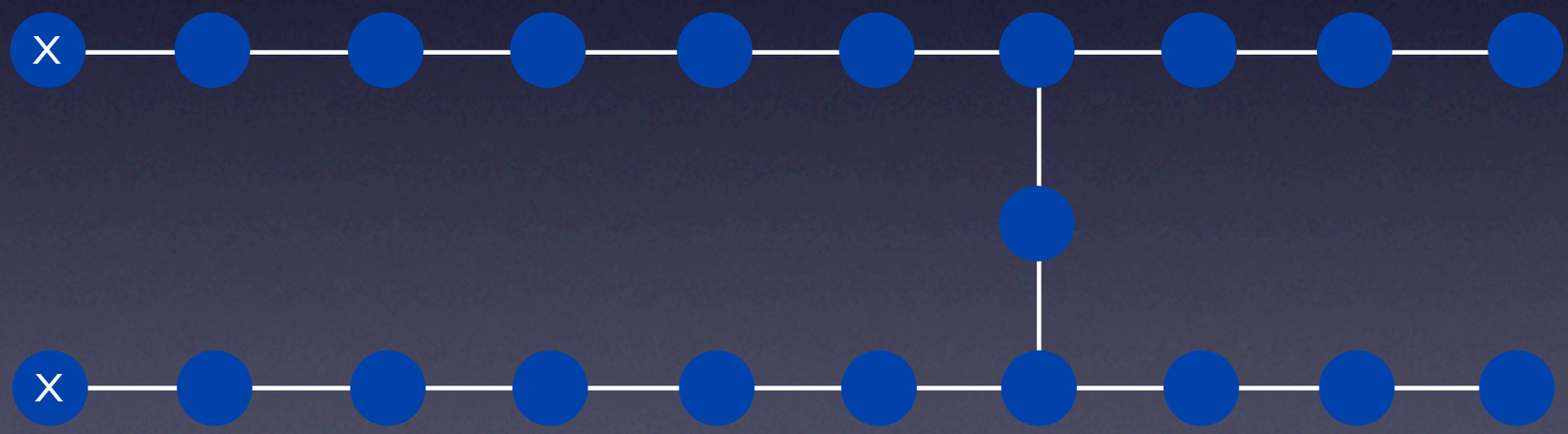
Create Entangled State

One Way QC



[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]



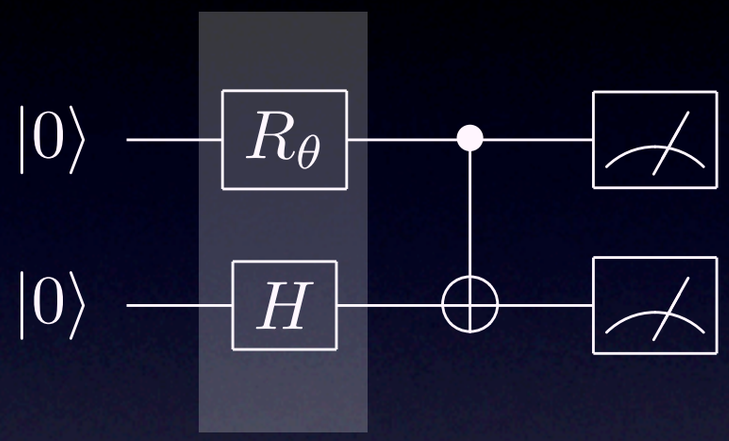


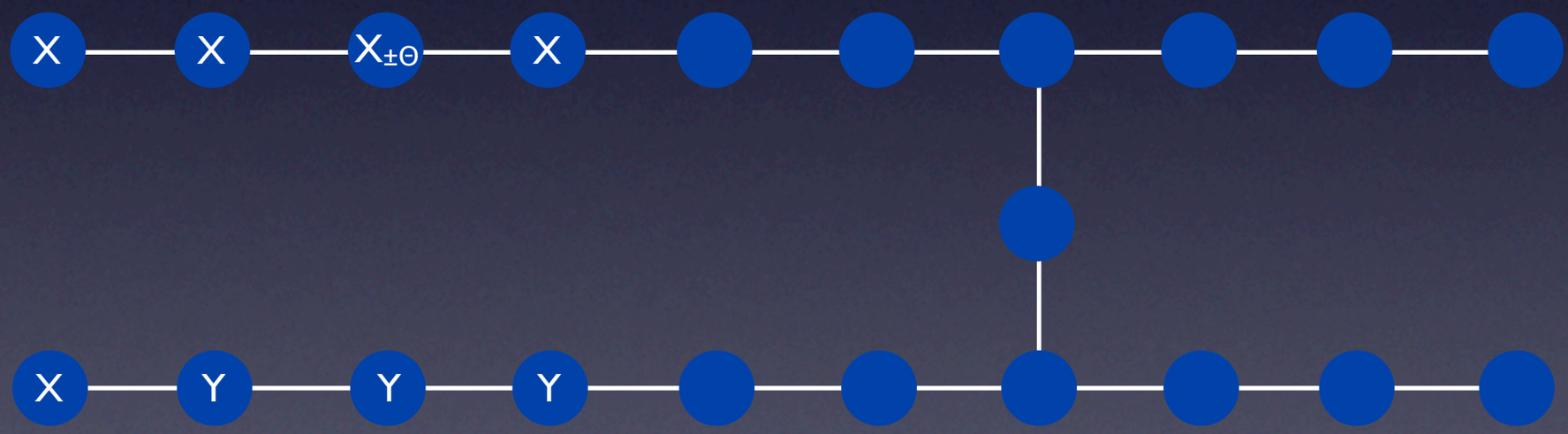
Adaptively measure to enact circuit

One Way QC



[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]



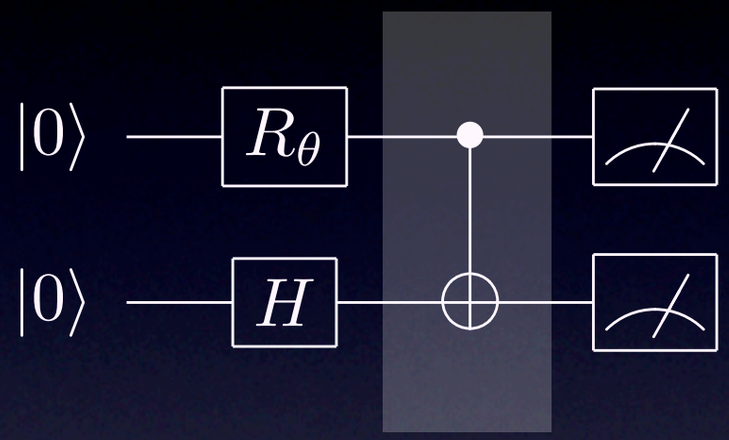


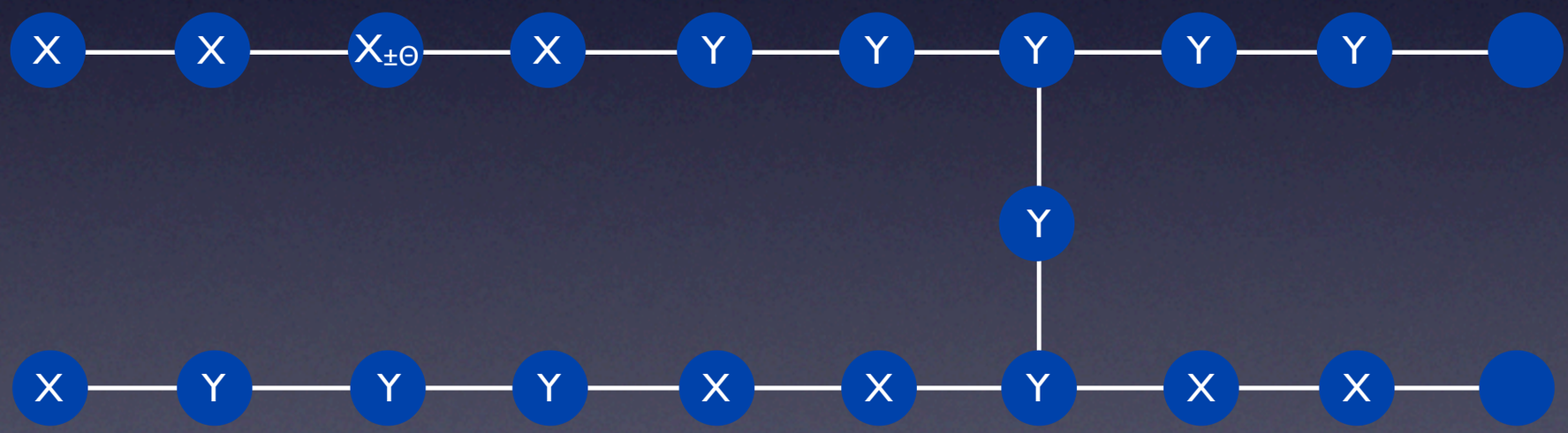
Adaptively measure to enact circuit

One Way QC



[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]



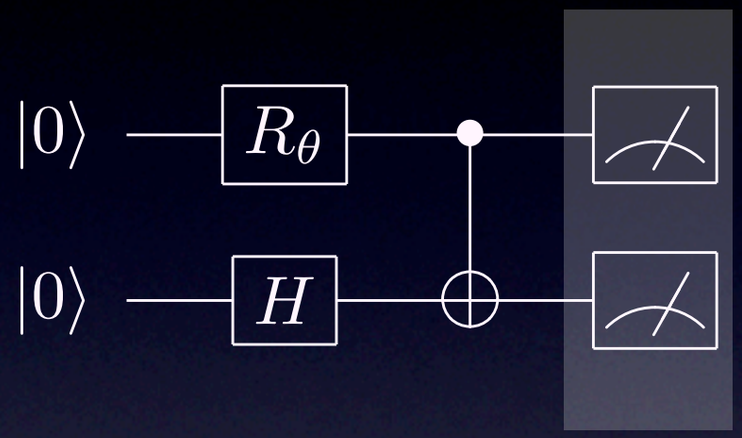


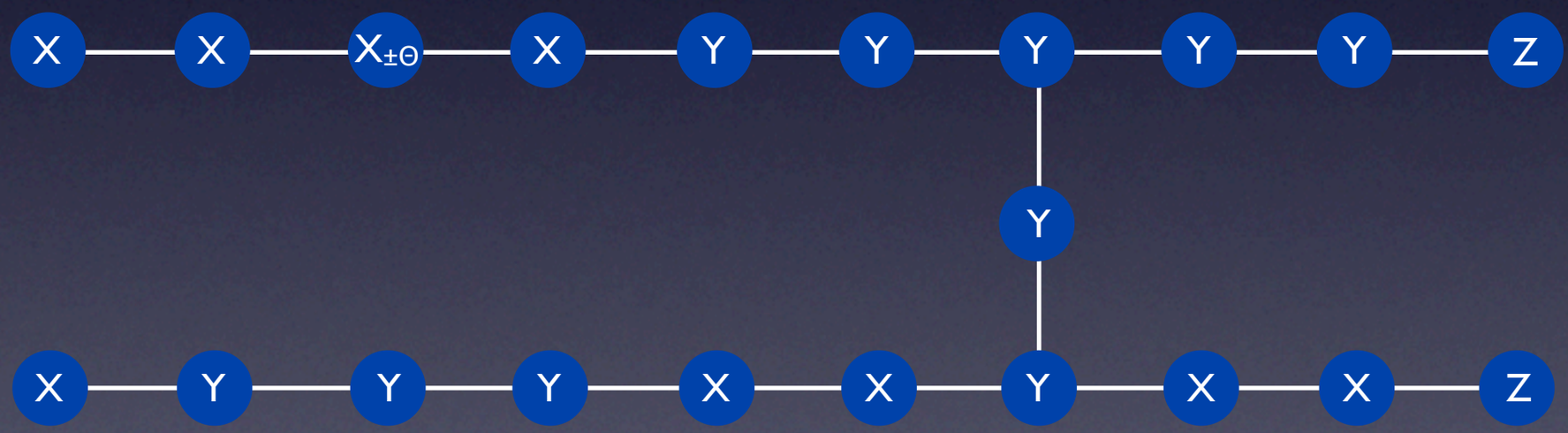
Adaptively measure to enact circuit

One Way QC



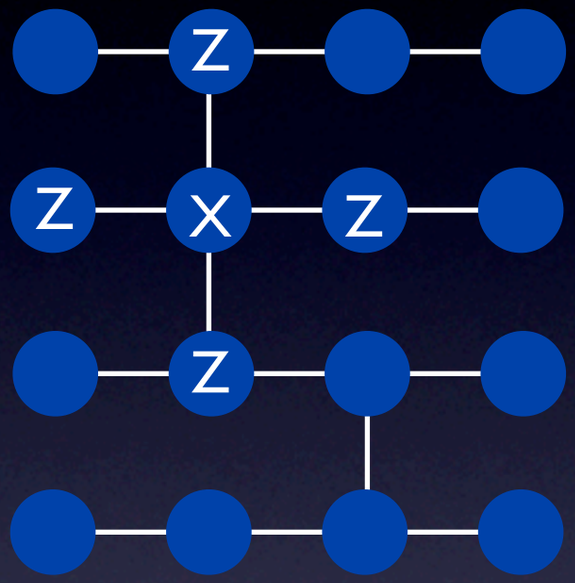
[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]





Adaptively measure to enact circuit

Cluster State Hamiltonian



$$S_v = X_v \prod_{w \text{ adjacent to } v} Z_w$$

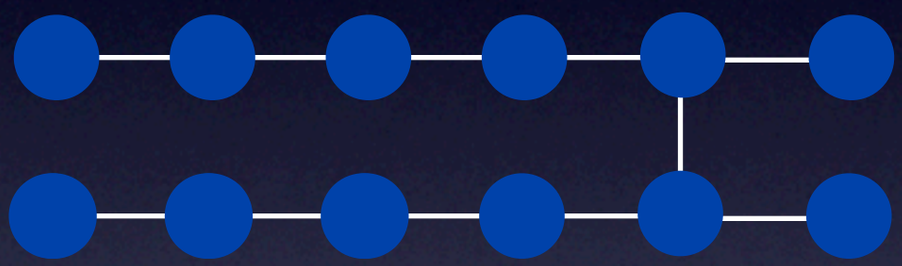
Cluster state is ground state of $H_C = -\Delta \sum_v S_v$

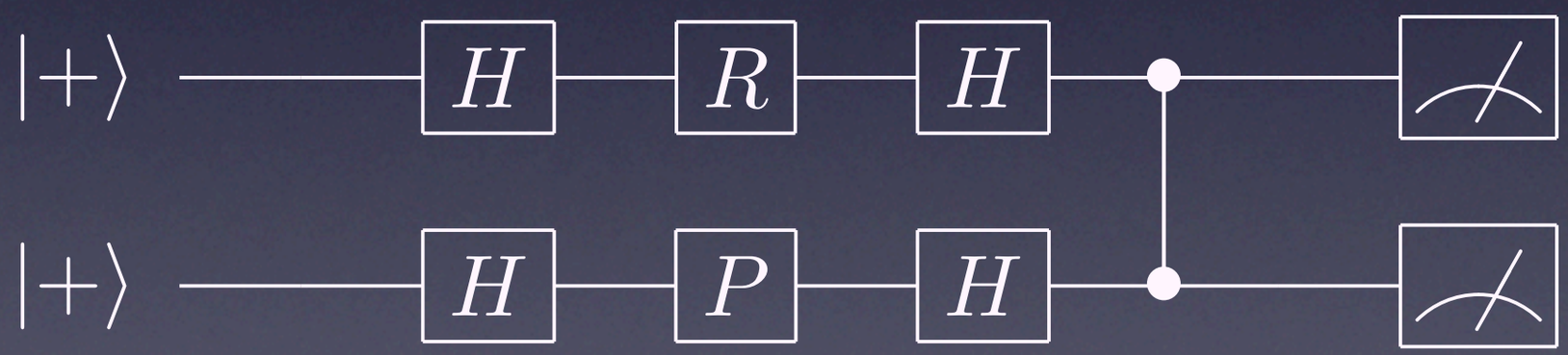
Possible to use gadgets to make only 2-qubit interactions:

[S. D. Bartlett and T. Rudolph, Phys. Rev. A 74, 040302 (2006)]

Adiabatic Cluster State QC

By turning on single qubit Hamiltonians while turning off parts of the cluster state Hamiltonian, we can enact a quantum circuit:

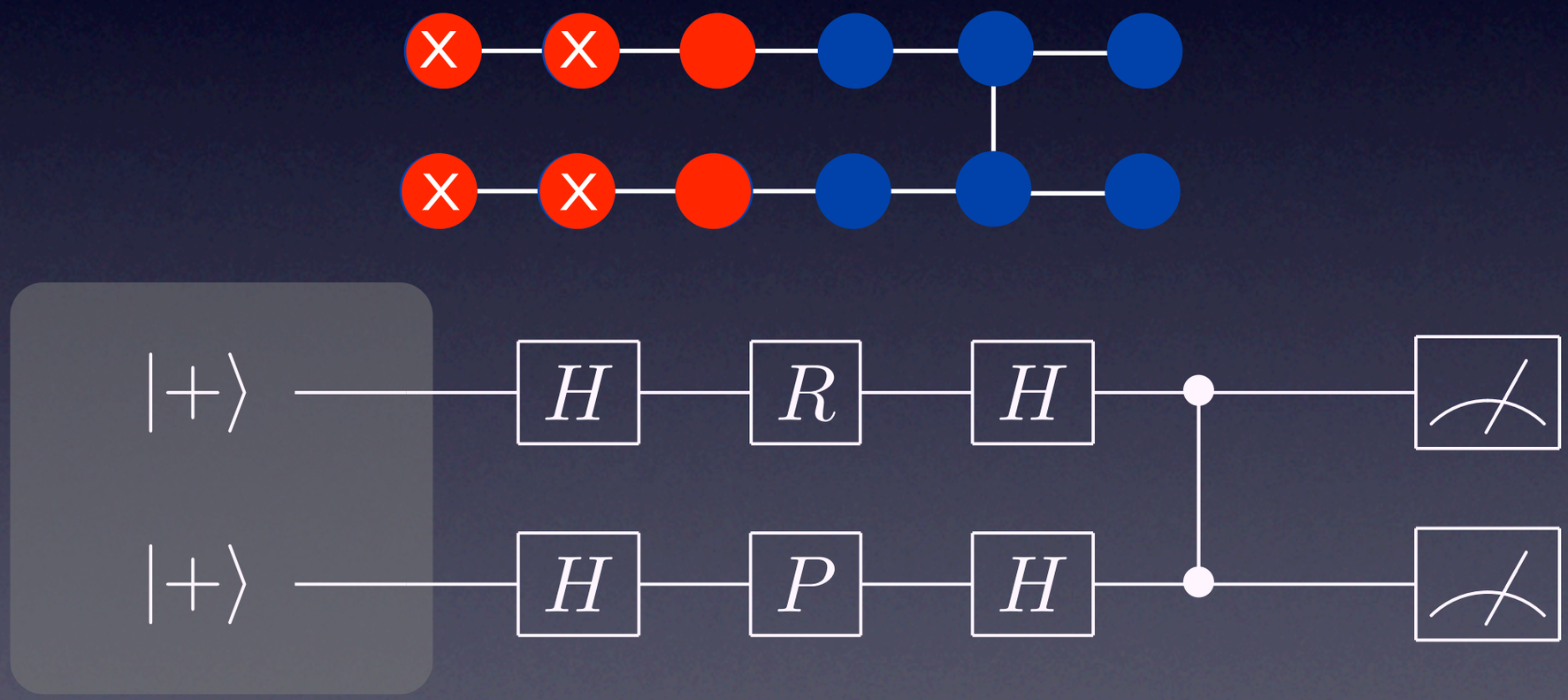




[Bacon and Flammia, "Adiabatic Cluster State Computing," arXiv:0912.2098 (2009)]

Adiabatic Cluster State QC

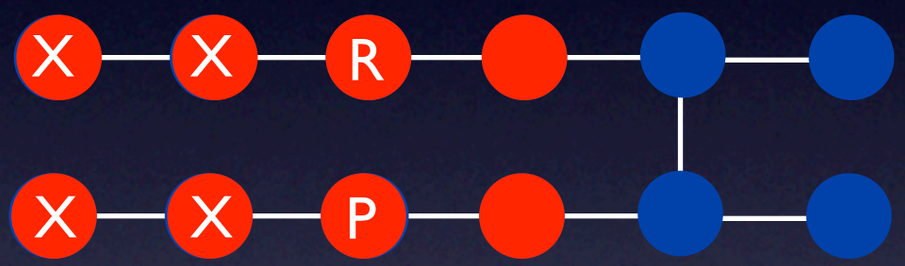
By turning on single qubit Hamiltonians while turning off parts of the cluster state Hamiltonian, we can enact a quantum circuit:

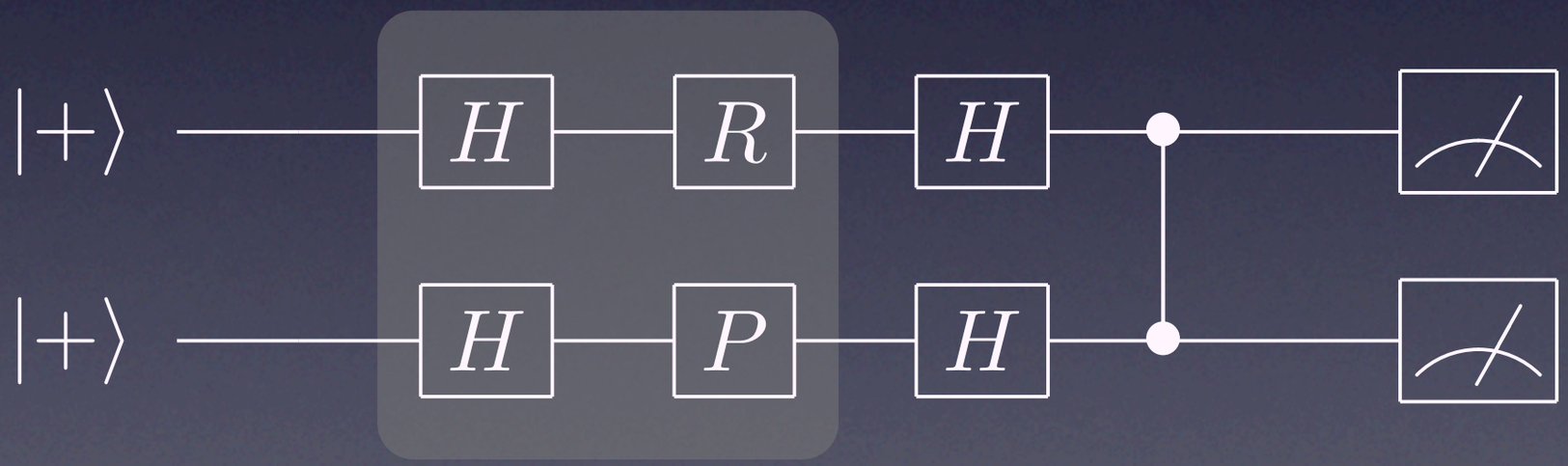


[Bacon and Flammia, "Adiabatic Cluster State Computing," arXiv:0912.2098 (2009)]

Adiabatic Cluster State QC

By turning on single qubit Hamiltonians while turning off parts of the cluster state Hamiltonian, we can enact a quantum circuit:

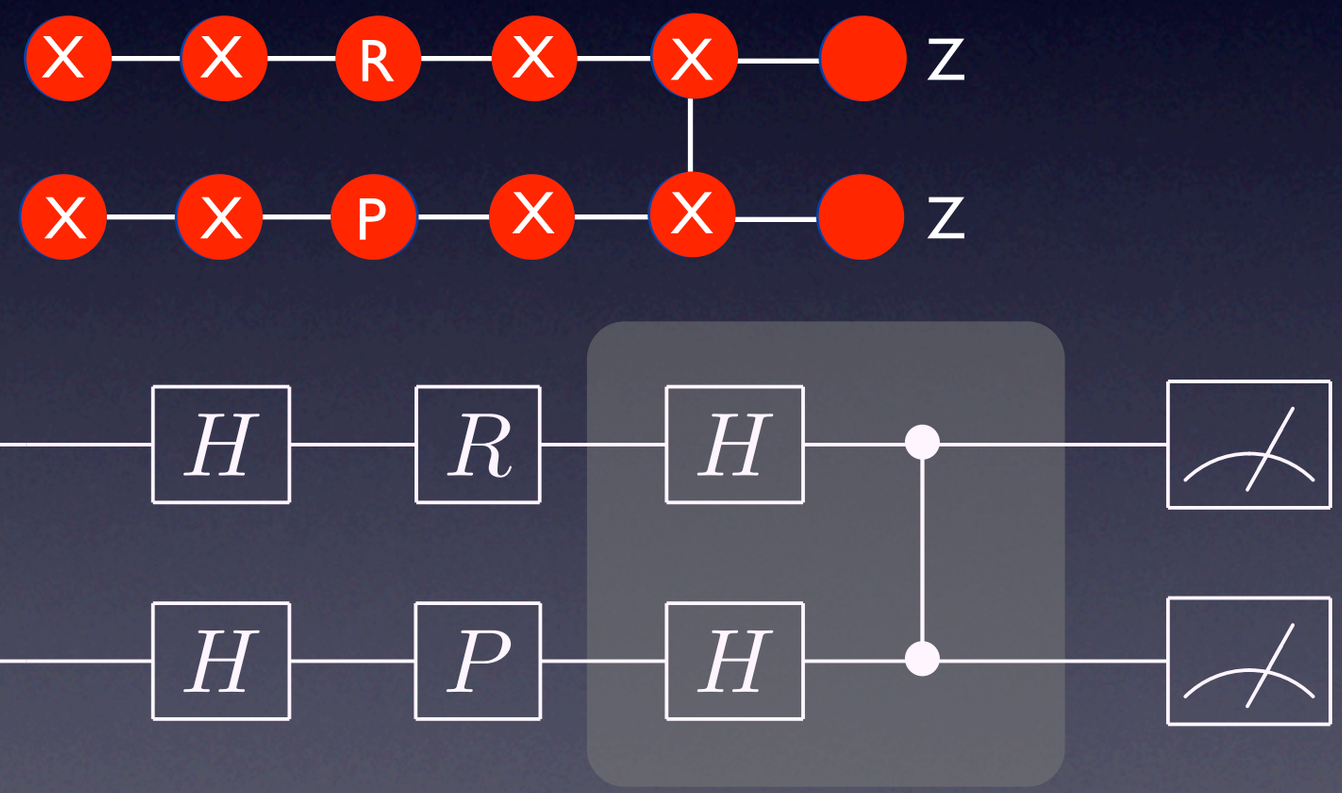




[Bacon and Flammia, "Adiabatic Cluster State Computing," arXiv:0912.2098 (2009)]

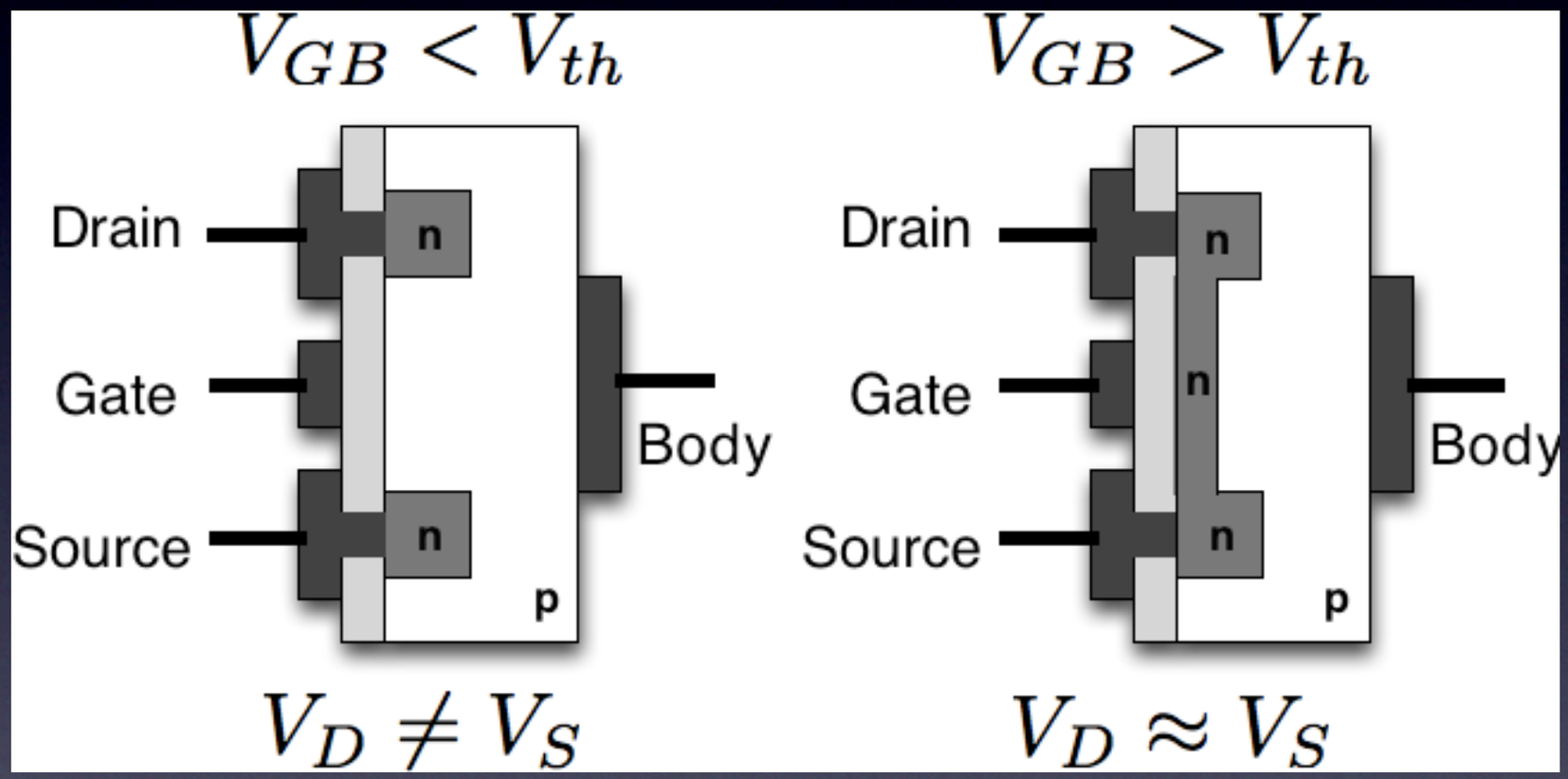
Adiabatic Cluster State QC

By turning on single qubit Hamiltonians while turning off parts of the cluster state Hamiltonian, we can enact a quantum circuit:



[Bacon and Flammia, "Adiabatic Cluster State Computing," arXiv:0912.2098 (2009)]

Can We Make a Classical Transistor?

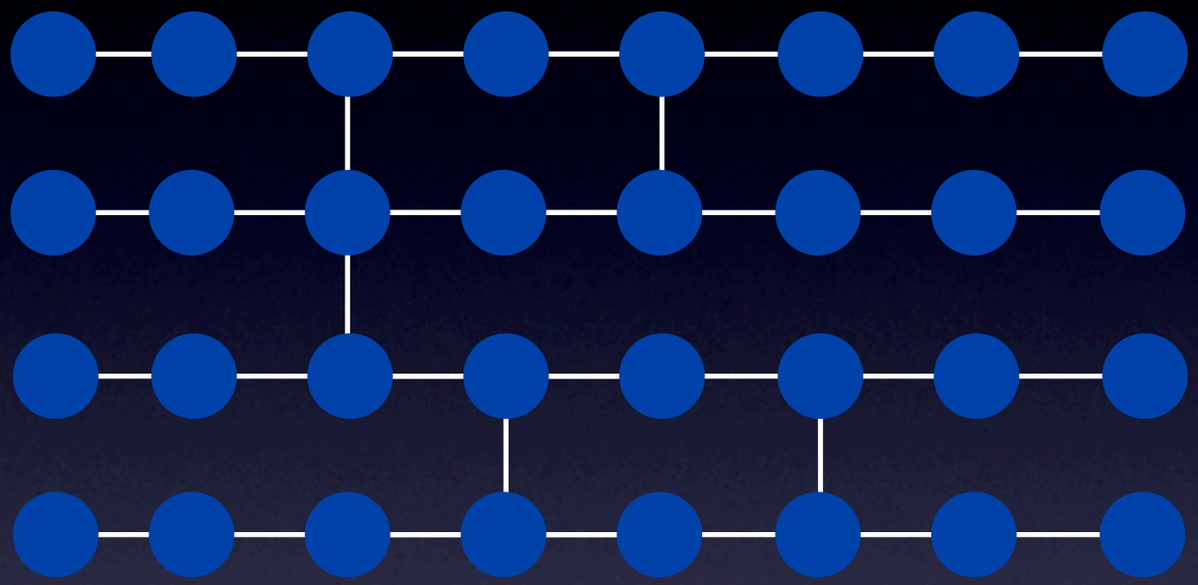


An "identity gate"

Problem: quantum information cannot be cloned

Quantum Transistor

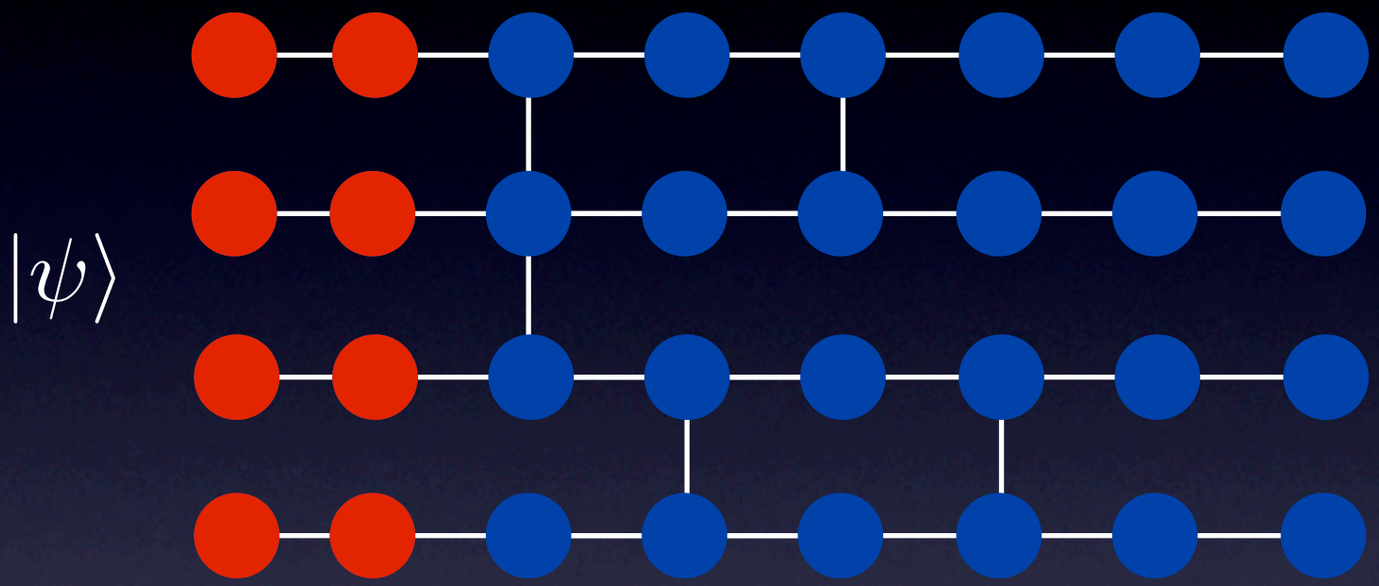
[Bacon, Crosswhite, Flammia "Adiabatic Quantum Transistors" (2010)
ask me for a copy]



1. Many-body system in its ground state

Quantum Transistor

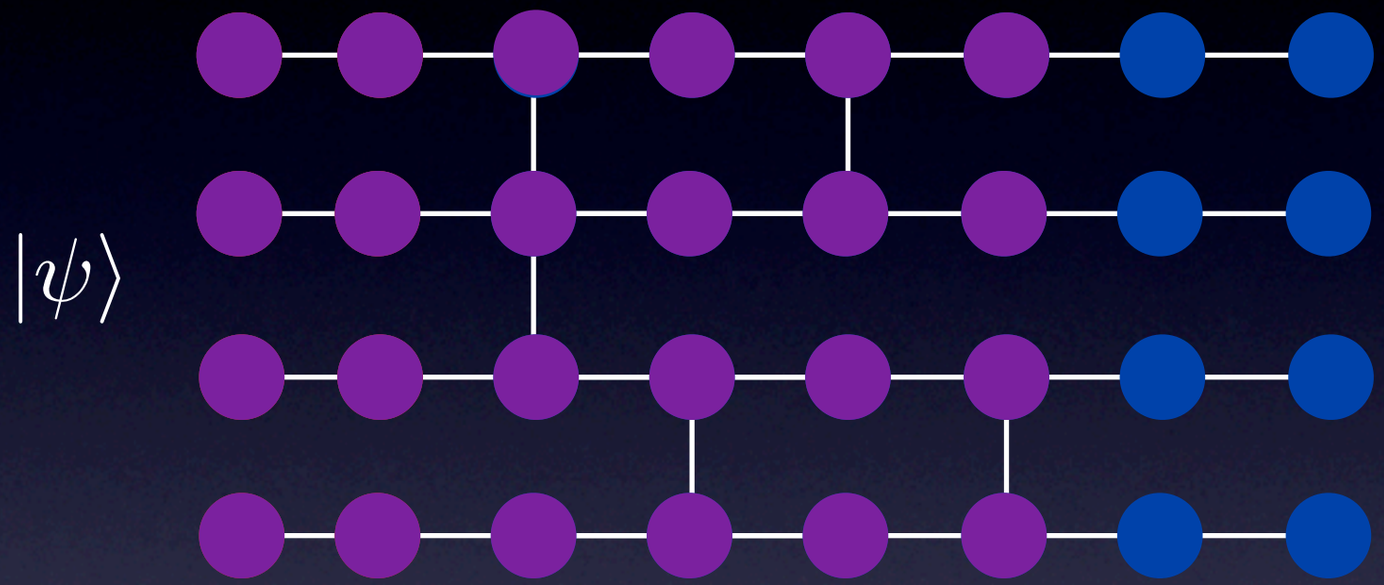
[Bacon, Crosswhite, Flammia "Adiabatic Quantum Transistors" (2010) ask me for a copy]



1. Many-body system in its ground state
2. Qubits localized on one side of the device

Quantum Transistor

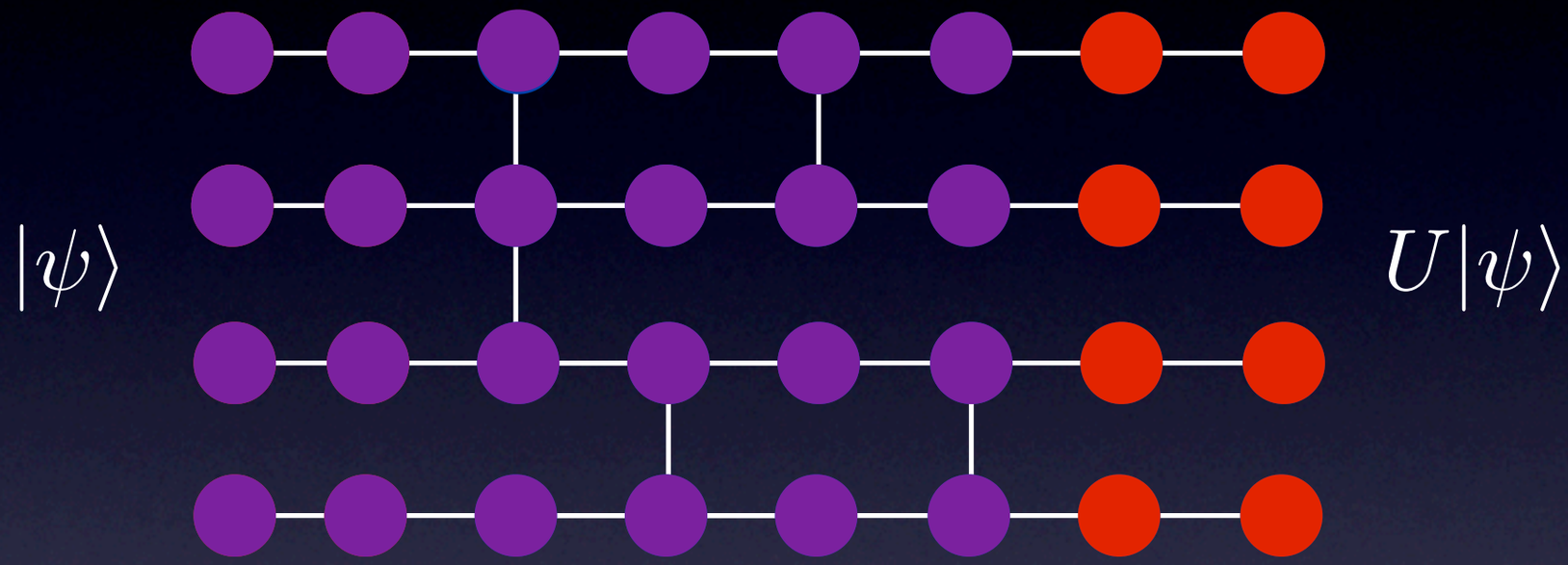
[Bacon, Crosswhite, Flammia "Adiabatic Quantum Transistors" (2010) ask me for a copy]



1. Many-body system in its ground state
2. Qubits localized on one side of the device
3. Apply a strong 1-qubit external field to device

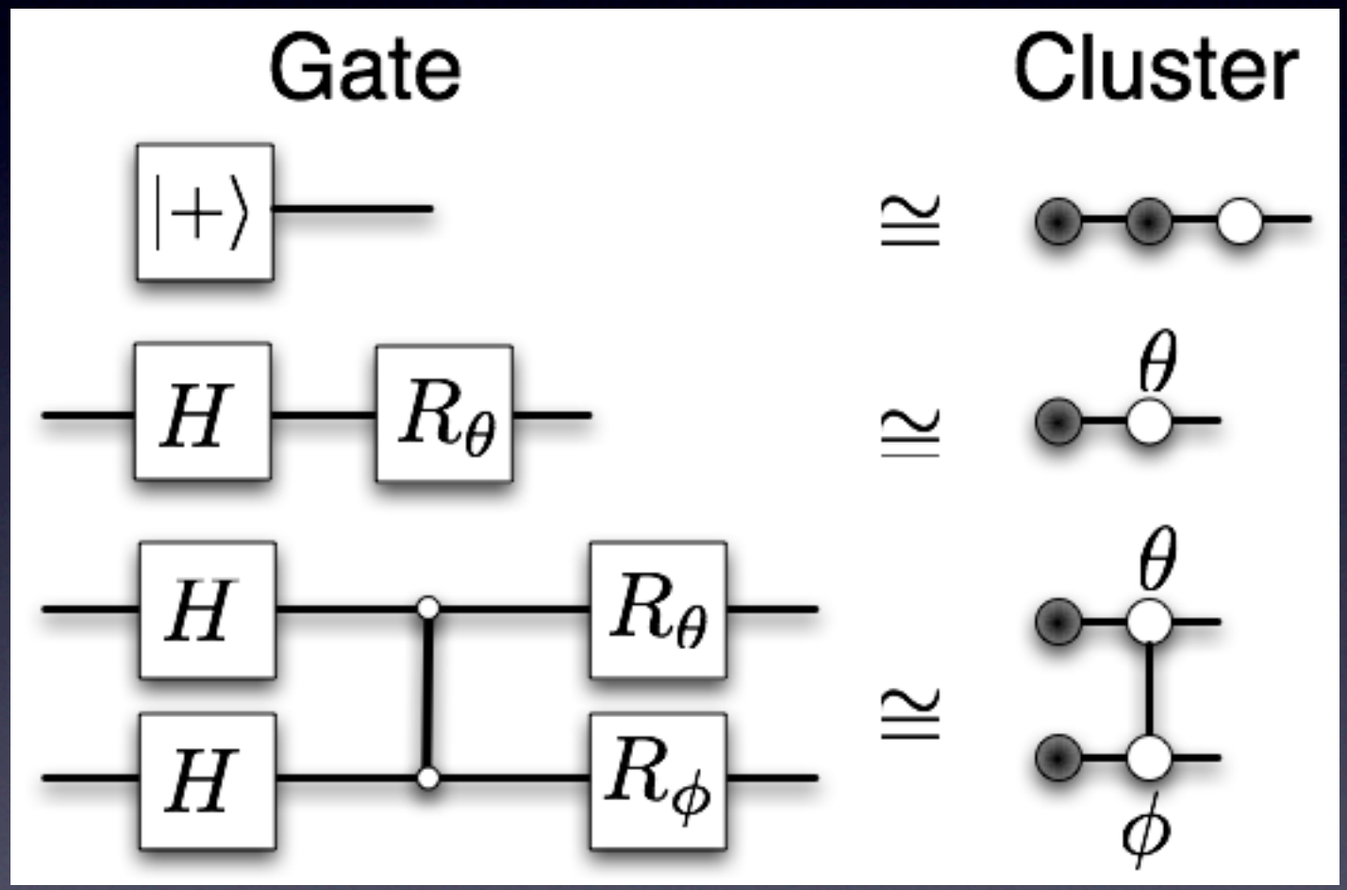
Quantum Transistor

[Bacon, Crosswhite, Flammia "Adiabatic Quantum Transistors" (2010) ask me for a copy]



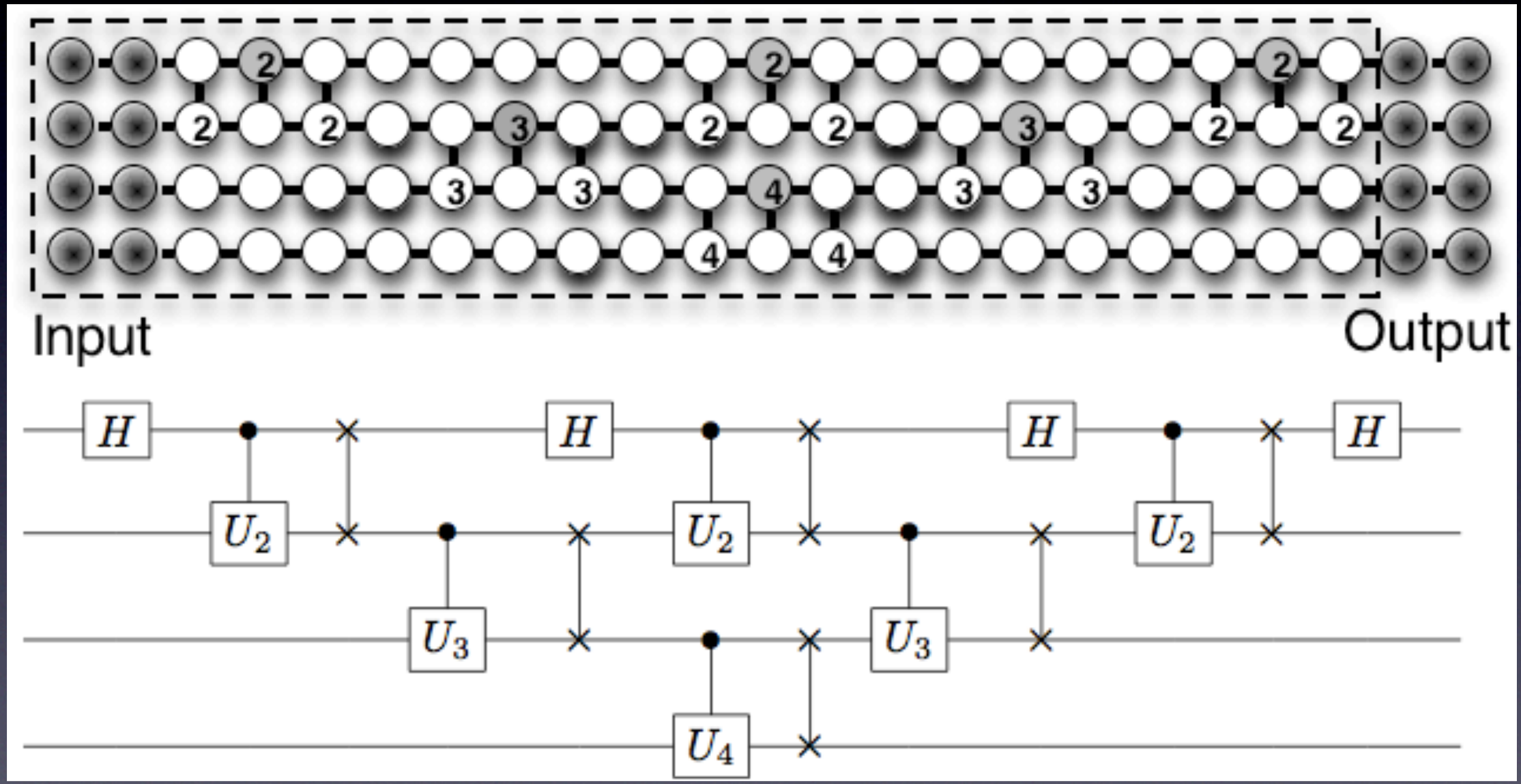
1. Many-body system in its ground state
2. Qubits localized on one side of the device
3. Apply a strong 1-qubit external field to device
4. Qubits now localized on other side of device with a quantum circuit applied to the qubits

Quantum Transistor Dictionary



$$R(\theta) = \exp(-i\theta Z/2)$$

Example

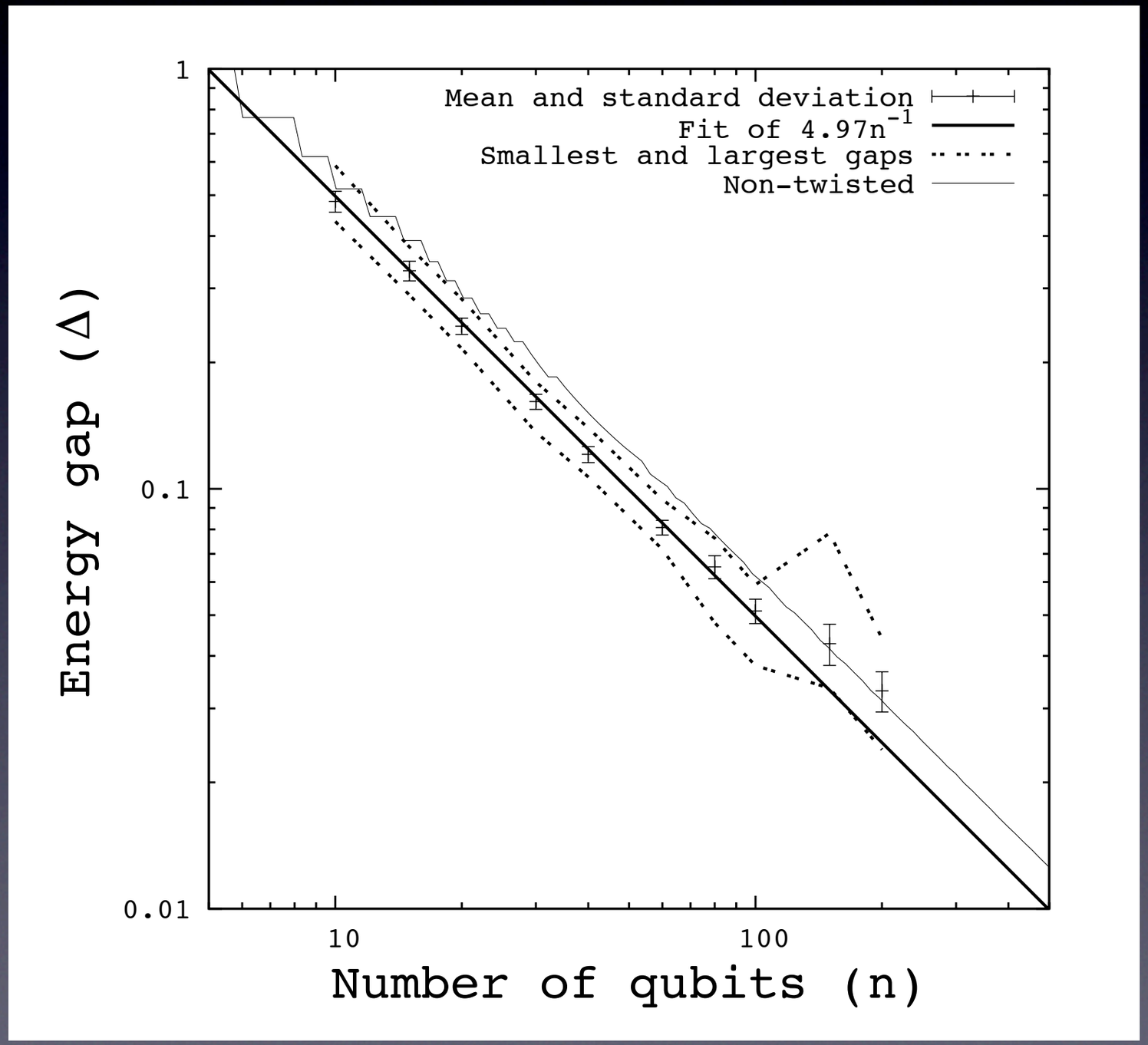


Adiabatic

- How slowly must we turn on the external field in order for the device to successfully quantum compute?

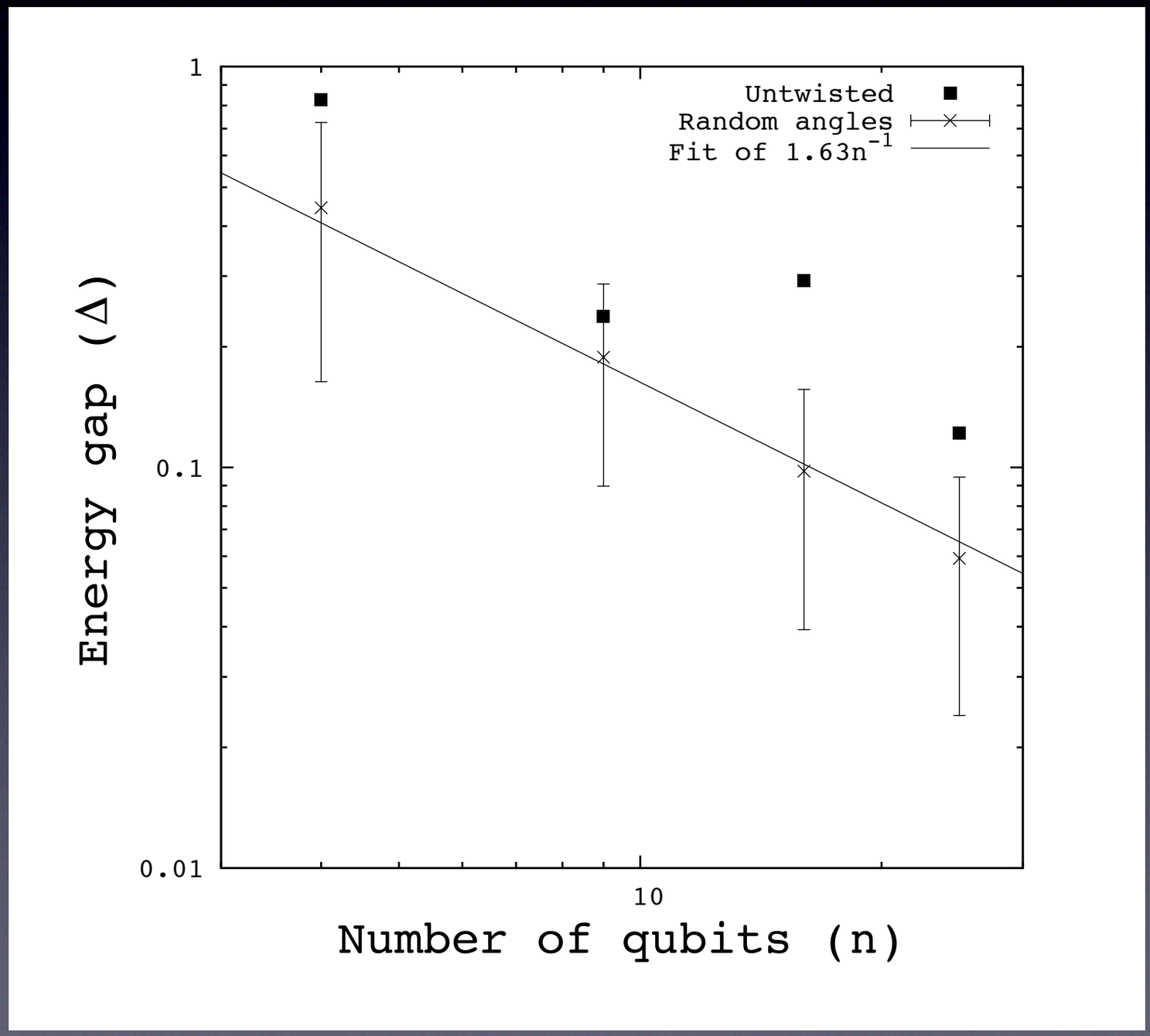
Adiabatic Quantum Transistor

Gap for 1D circuit with random angle rotations...



Adiabatic Quantum Transistor

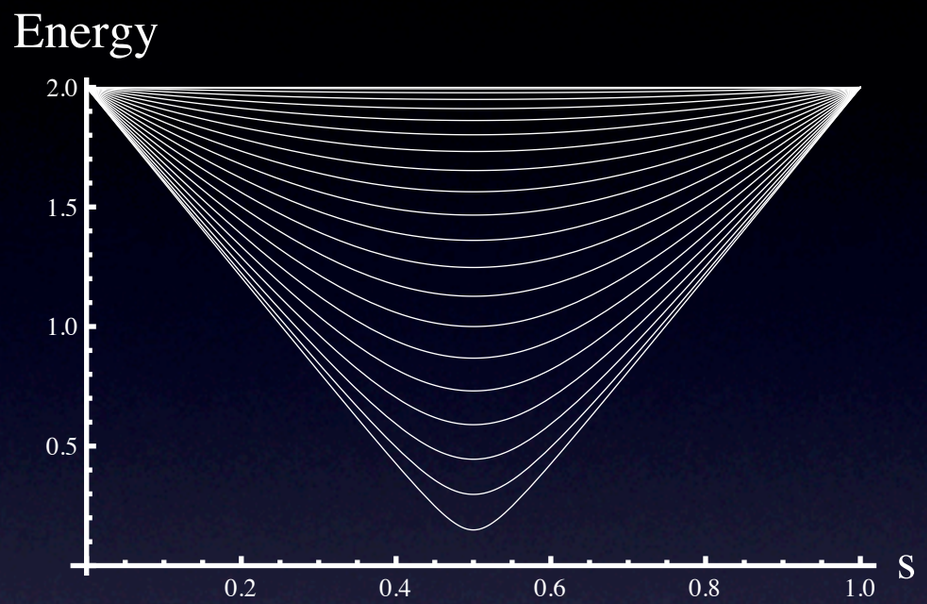
Gap for 2D circuit with random angle rotations...



Adiabatic

- How slowly must we turn on the external field in order for the device to successfully quantum compute?
  - 1D no twists, proven polynomial in size of circuit
  - 1D twists, extremely strong polynomial in size of circuit
  - 2D twists, evidence of polynomial in size of circuit

Fault-Tolerance?



For 1D untwisted Hamiltonian, decouples into 2 1D Ising chains with transverse field.

Two types of errors:

- Those that change energy
- Those that act on degeneracy

Which lead to

- Errors that change energy can be mapped to circuit model errors (independent error model)
- Degeneracy errors only don't cost energy at small and large times. Adjust schedule to make independent

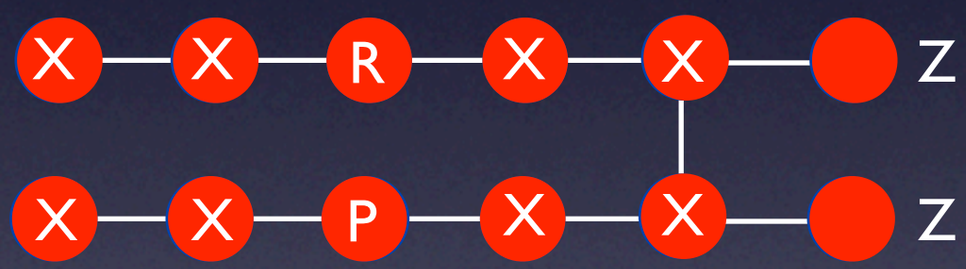
1D errors mapped to independent error model

Summary

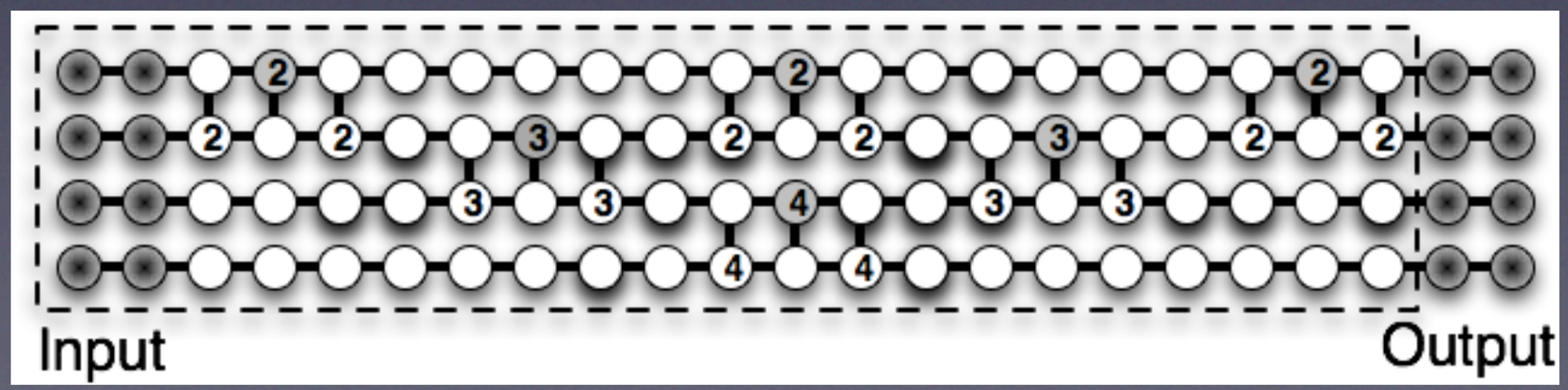
- Adiabatic teleportation



- Adiabatic cluster state computing



- Adiabatic quantum transistors



QW Group

Dave Bacon

<u>Graduate students</u>

Lukas Svec (physics)

Jijian Yan (physics)

Gregory Crosswhite (physics)

Isaac Crosson (physics)

Kamil Michnicki (physics)

Paul Pham (CS)

Amplifying gadgets

Fault-tolerant bang-bang simulation

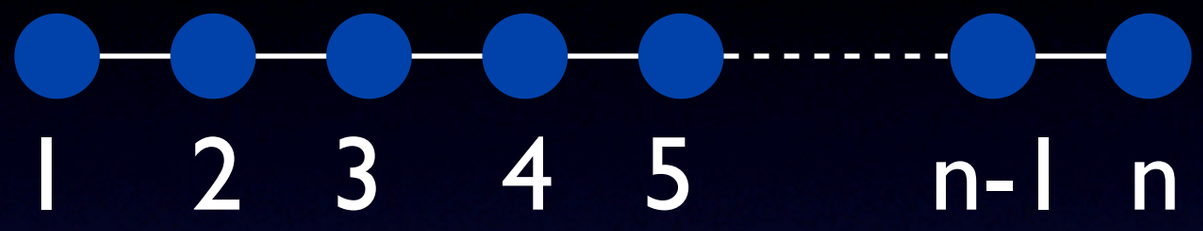
CodeQuest

Adiabatic Transistors

Fault-tolerant classical ground state computing

Adiabatic Teleportation

One Qubit



Modified cluster Hamiltonian:

$$H_C = -\Delta \sum_{i=1}^{n-1} S_i \qquad S_i = [Z]_i[X]_{i+1}[Z]_{i+2} \quad 1 \leq i \leq n-2$$

$$S_{n-1} = [Z]_{n-1}[X]_n$$

Ground state is 2-fold degenerate, can define logical operators for this qubit:

$$S_1 \rightarrow [X]_1$$

$$\bar{X} = [X]_1[Z]_2 \qquad \bar{X} \cong [Z]_2$$

$$\bar{Z} = [Z]_1 \qquad \bar{Z} \cong [X]_2[Z]_3$$